SAP Bug Tracking Management System — Full Implementation Guide

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HƯỚNG DẪN TRIỂN KHAI CHI TIẾT — PHASE A: DATABASE HARDENING

Dự án: SAP Bug Tracking Management System **Ngày:** 24/03/2026 | **Phiên bản:** 5.0 (Module Pool Integration) **Thời gian ước tính:** 1 ngày (24-25/03) **Development Account:** `DEV-089` (Pass: `@Anhtuoi123`) — *Quyền SE11, SE80*

Bước A1: Tạo Domains & Data Elements mới

Mục tiêu: Tạo các Domain và Data Element cần thiết cho Project, Severity, Soft Delete.

Vào T-code **SE11**, tạo lần lượt:

1. Domains

Domain	Type	Length	Fixed Values
`ZDOM_PROJECT_ID`	CHAR	20	
`ZDOM_PRJ_STATUS`	CHAR	1	1=Opening, 2=In-Process, 3=Done, 4=Cancel
`ZDOM_SEVERITY`	CHAR	1	1=Dump, 2=Very-High, 3=High, 4=Normal, 5=Minor
`ZDOM_IS_DEL`	CHAR	1	(space)=Active, X=Deleted

Cách tạo Domain (ví dụ ZDOM_SEVERITY):

1. **SE11** → nhập `ZDOM_SEVERITY` → **Create**
2. **Short Description:** "Severity Level"
3. Tab **Definition:** Data Type = `CHAR`, No. Characters = `1`
4. Tab **Value Range** → Fixed Values:
   - `1` = Dump
   - `2` = Very High
   - `3` = High
   - `4` = Normal
   - `5` = Minor
5. **Save** → **Activate**
6. Lặp lại cho các Domain còn lại.

2. Data Elements

Data Element	Domain	Short Text
`ZDE_PROJECT_ID`	`ZDOM_PROJECT_ID`	Project ID
`ZDE_PRJ_NAME`	`CHAR100`	Project Name
`ZDE_PRJ_DESC`	`CHAR255`	Project Description
`ZDE_PRJ_STATUS`	`ZDOM_PRJ_STATUS`	Project Status
`ZDE_SEVERITY`	`ZDOM_SEVERITY`	Bug Severity Level
`ZDE_IS_DEL`	`ZDOM_IS_DEL`	Soft Delete Flag

Cách tạo Data Element (ví dụ ZDE_SEVERITY):

1. **SE11** → nhập `ZDE_SEVERITY` → **Create**
2. **Short Description:** "Bug Severity Level"
3. Tab **Data Type:** Domain = `ZDOM_SEVERITY`
4. Tab **Field Label:**
   - Short: Severity
   - Medium: Severity Level
   - Long: Bug Severity Level
   - Heading: Severity
5. **Save** → **Activate**
6. Lặp lại cho các Data Elements còn lại.

✅ **Checkpoint:** Tất cả Domains và Data Elements ở trạng thái Active.

Bước A2: Tạo bảng ZBUG_PROJECT

Mục tiêu: Bảng quản lý Project — mỗi Bug sẽ thuộc về 1 Project.

Vào **SE11** → nhập `ZBUG_PROJECT` → **Create** → **Transparent Table**.

Tab Fields

Field	Data Element	Type	Length	Key	Initial	Not Null
`MANDT`	`MANDT`	CLNT	3	✓		
`PROJECT_ID`	`ZDE_PROJECT_ID`	CHAR	20	✓		
`PROJECT_NAME`	`ZDE_PRJ_NAME`	CHAR	100			
`DESCRIPTION`	`ZDE_PRJ_DESC`	CHAR	255			
`START_DATE`	`SYDATUM`	DATS	8			
`END_DATE`	`SYDATUM`	DATS	8			
`PROJECT_MANAGER`	`ZDE_USERNAME`	CHAR	12			
`PROJECT_STATUS`	`ZDE_PRJ_STATUS`	CHAR	1		X	
`NOTE`	`CHAR255`	CHAR	255			
`ERNAM`	`ERNAM`	CHAR	12			
`ERDAT`	`ERDAT`	DATS	8			
`ERZET`	`ERZET`	TIMS	6			
`AENAM`	`AENAM`	CHAR	12			
`AEDAT`	`AEDAT`	DATS	8			
`AEZET`	`AEZET`	TIMS	6			
`IS_DEL`	`ZDE_IS_DEL`	CHAR	1		X	

Lưu ý: Field `PROJECT_STATUS` default = `1` (Opening). Field `IS_DEL` default = (Active).

Tab Technical Settings

- **Data Class:** `APPL0` (Master Data)
- **Size Category:** `0` (0 - 2,600 records)
- **Buffering:** Not allowed

Tab Enhancement Category

- Can Be Enhanced (Character-Type or Numeric)

Nhấn **Save** → **Activate**.

✅ **Checkpoint: SE16** → `ZBUG_PROJECT` → hiển thị bảng rỗng, cấu trúc đúng 16 fields.

Bước A3: Tạo bảng ZBUG_USER_PROJEC

Mục tiêu: Bảng liên kết Many-to-Many giữa User và Project.

Vào **SE11** → nhập `ZBUG_USER_PROJEC` → **Create** → **Transparent Table**.

Tab Fields

Field	Data Element	Type	Length	Key
`MANDT`	`MANDT`	CLNT	3	✓
`USER_ID`	`ZDE_USERNAME`	CHAR	12	✓
`PROJECT_ID`	`ZDE_PROJECT_ID`	CHAR	20	✓
`ROLE`	`ZDE_BUG_ROLE`	CHAR	1	
`ERNAM`	`ERNAM`	CHAR	12	
`ERDAT`	`ERDAT`	DATS	8	
`ERZET`	`ERZET`	TIMS	6	
`AENAM`	`AENAM`	CHAR	12	
`AEDAT`	`AEDAT`	DATS	8	
`AEZET`	`AEZET`	TIMS	6	

> ⚠️ **Lưu ý ROLE field:** - Data Element `ZDE_BUG_ROLE` đã có sẵn trong bảng `ZBUG_USERS` (CHAR 1, values T=Tester, D=Developer, M=Manager) - Đây là **cùng một data element** — user có thể có role khác nhau trong từng project - **Không thêm ROLE vào ZBUG_USERS** (nó đã có sẵn ở đó từ đầu) - Nếu SAP báo `ZDE_BUG_ROLE not found` khi tạo bảng → data element này đang được define trong domain `ZDOM_BUG_ROLE`; kiểm tra SE11 → ZBUG_USERS để xem data element thực tế đang dùng cho field ROLE

Tab Technical Settings

- **Data Class:** `APPL0`
- **Size Category:** `0`

Nhấn **Save** → **Activate**.

> ✅ **Checkpoint: SE16** → `ZBUG_USER_PROJEC` → hiển thị bảng rỗng, 3 key fields + field ROLE.

Bước A4: Cập nhật bảng ZBUG_TRACKER

Mục tiêu: Thêm `PROJECT_ID`, `SEVERITY`, Audit Fields, `IS_DEL` vào bảng bug hiện có.

> ⚠️ **QUAN TRỌNG — Audit Fields:**

Bảng ZBUG_TRACKER hiện tại (từ developer-guide.md) chỉ có `CREATED_AT` (DATS) và `CREATED_TIME` (TIMS) — đây KHÔNG phải ERNAM/ERDAT/ERZET chuẩn SAP.

- `ERNAM/ERDAT/ERZET` là **fields MỚI** (Created by/date/time theo SAP convention)
- `AENAM/AEDAT/AEZET` cũng là **fields MỚI** (Last changed by/date/time)
- Giữ lại `CREATED_AT/CREATED_TIME` cũ (không xóa, tránh mất data), nhưng code mới sẽ dùng `ERNAM/ERDAT/ERZET`

Sau khi thêm fields, cần **backfill** data cũ:

```
UPDATE zbug_tracker SET ernam = tester_id, erdat = created_at, erzet =
created_time
  WHERE ernam IS INITIAL.
```

Vào **SE11** → nhập `ZBUG_TRACKER` → **Change**.

Thêm các trường sau vào cuối bảng:

Field	Data Element	Chức năng	Ghi chú
`PROJECT_ID`	`ZDE_PROJECT_ID`	FK → ZBUG_PROJECT, Bug thuộc project nào	
`SEVERITY`	`ZDE_SEVERITY`	1=Dump, 2=VHigh, 3=High, 4=Normal, 5=Minor	Default '4'
`VERIFY_TESTER_ID`	`ZDE_USERNAME`	Tester verify fix (khi status → Resolved)	
`APPROVED_BY`	`ZDE_USERNAME`	Manager approve close	
`APPROVED_AT`	`DATS`	Ngày approve close	
`CLOSED_AT`	`DATS`	Ngày đóng bug (auto fill khi status → 7)	
`ERNAM`	`ERNAM`	Created by (SAP standard)	**MỚI** — backfill từ TESTER_ID
`ERDAT`	`ERDAT`	Created date (SAP standard)	**MỚI** — backfill từ CREATED_AT
`ERZET`	`ERZET`	Created time (SAP standard)	**MỚI** — backfill từ CREATED_TIME
`AENAM`	`AENAM`	Last changed by	**MỚI**
`AEDAT`	`AEDAT`	Last changed date	**MỚI**
`AEZET`	`AEZET`	Last changed time	**MỚI**
`IS_DEL`	`ZDE_IS_DEL`	Soft delete flag ('X' = deleted)	

Lưu ý:

- Dùng `ZDE_USERNAME` (CHAR 12) cho `VERIFY_TESTER_ID` và `APPROVED_BY` — nhất quán với `TESTER_ID/DEV_ID` đã có trong bảng
- SEVERITY default nên là '4' (Normal)
- `CLOSED_AT` tự động fill khi status chuyển sang 7 (Closed) trong FM `Z_BUG_UPDATE_STATUS`
- `VERIFY_TESTER_ID` fill khi Tester verify pass (status 5→6)

Nhấn **Save** → **Activate**.

⚠️ **NẾU bảng đã có dữ liệu:**

- SAP sẽ yêu cầu Adjust Database
- Vào **SE14** → nhập `ZBUG_TRACKER` → **Adjust**
- Chọn "Activate and adjust database table"
- Kiểm tra log, đảm bảo không mất dữ liệu

Backfill audit fields cho dữ liệu cũ (chạy 1 lần trong SE38):

```
REPORT z_bug_backfill_audit.
" Backfill ERNAM/ERDAT/ERZET từ CREATED_AT/CREATED_TIME/TESTER_ID
UPDATE zbug_tracker
  SET ernam = tester_id
      erdat = created_at
      erzet = created_time
  WHERE ernam = '' OR ernam IS INITIAL.
COMMIT WORK.
WRITE: / 'Backfill complete. Records updated:', sy-dbcnt.
```

✅ **Checkpoint: SE11** → `ZBUG_TRACKER` → Display → thấy `PROJECT_ID`, `SEVERITY`, `ERNAM`, `ERDAT`, `ERZET`, `IS_DEL` (tổng cộng +13 fields mới).

Bước A5: Cập nhật bảng ZBUG_USERS

Mục tiêu: Thêm Audit Fields, `IS_DEL`, Email OBLIGATORY.

Vào **SE11** → nhập `ZBUG_USERS` → **Change**.

Thêm các trường (nếu chưa có):

Field	Data Element	Chức năng
`AENAM`	`AENAM`	Last changed by
`AEDAT`	`AEDAT`	Last changed date
`AEZET`	`AEZET`	Last changed time
`IS_DEL`	`ZDE_IS_DEL`	Soft delete flag

Cập nhật field EMAIL:

1. Click vào field `EMAIL`
2. Nếu dùng Data Element → vào Data Element → sửa Domain → Check "Required Entry" / "Obligatory"
3. Hoặc trực tiếp trong bảng: check column "Not Null" cho field `EMAIL`

Nhấn **Save** → **Activate** → **SE14** Adjust nếu cần.

> ✅ **Checkpoint: SE16** → `ZBUG_USERS` → cấu trúc có `AENAM`, `AEDAT`, `AEZET`, `IS_DEL`.

Bước A6: Tạo Message Class ZBUG_MSG

Mục tiêu: Tạo Message Class để hỗ trợ đa ngôn ngữ (EN/VI).

Vào **SE91** → nhập `ZBUG_MSG` → **Create**.

- **Short Text:** "Bug Tracking System Messages"

Tab **Messages** — tạo các message số sau (EN trước):

No.	Short Text (EN)
000	&1 &2 &3 &4 (Generic placeholder)
001	Bug &1 created successfully
002	Bug &1 updated successfully
003	User not found in system
004	User is not a member of this project
005	Only Tester can create bugs
006	Not authorized to update this bug
007	Only Manager can delete projects
008	Project must be Done or Cancel to delete
009	Please select a row first
010	Database insert failed
011	Failed to generate Bug ID
012	Project &1 created successfully
013	Project still has unresolved bugs
014	Employee cannot work on multiple projects at same time
015	Upload BUGPROOF file before creating bug
016	Upload TESTCASE file before marking as Fixed
017	Upload CONFIRM file before marking as Resolved
018	Invalid status transition from &1 to &2
019	Are you sure you want to delete this &1?
020	Operation cancelled
021	Project &1 does not exist or is inactive
022	Bug can only be created in InProcess projects
023	Email is required for user registration
024	Invalid email format
025	Successfully saved
026	Data refreshed
027	Email sent successfully
028	Status changed from &1 to &2
029	User &1 does not exist or is inactive
030	High severity bug must have High priority
031	Bug &1 reassigned to &2 successfully
032	New developer must belong to the same project

Nhấn **Save** → **Activate**.

Thêm bản dịch Tiếng Việt:

1. Vẫn ở **SE91** → `ZBUG_MSG` → chọn **Goto** → **Translation**
2. Hoặc dùng T-code **SE63** → Short Texts → SE91 Messages
3. Target Language: **VI** (Vietnamese)
4. Dịch từng message, ví dụ:
   - `001`: "Bug &1 đã tạo thành công"
   - `003`: "Không tìm thấy user trong hệ thống"
   - `005`: "Chỉ Tester mới được tạo bug"
   - `031`: "Bug &1 đã chuyển giao cho &2 thành công"
5. **Save**

Cách sử dụng Message Class trong code:

```
" Thay vì hardcode:
" ev_message = 'User not found in system'.

" Dùng Message Class:
MESSAGE s003(zbug_msg) INTO ev_message.
" hoặc
MESSAGE s001(zbug_msg) WITH ev_bug_id INTO ev_message.
" hoặc hiển thị trực tiếp (popup/status bar):
MESSAGE s026(zbug_msg).  " 'Data refreshed'
MESSAGE s018(zbug_msg) WITH lv_old_status lv_new_status DISPLAY LIKE 'E'.
```

> ✅ **Checkpoint: SE91** → `ZBUG_MSG` → thấy 33+ messages, chuyển logon language EN/VI → text thay đổi.

Bước A7: Tạo Text Object ZBUG_NOTE (Long Text)

Mục tiêu: Tạo Text Object để lưu Long Text cho Dev Note, Tester Note, Root Cause.

Vào **SE75** → **Create**:

Text Object: `ZBUG`

- **Description:** "Bug Tracking Long Texts"

Tạo 3 Text IDs:

Text ID	Description	Text Format
`Z001`	Developer Note	*
`Z002`	Tester/Functional Note	*
`Z003`	Root Cause Analysis	*

Cách đọc / ghi Long Text bằng ABAP:

```
" --- Đọc Long Text ---
DATA: lt_lines TYPE TABLE OF tline,
      ls_header TYPE thead.

ls_header-tdname    = lv_bug_id.       " Object key = Bug ID
ls_header-tdid      = 'Z001'.          " Text ID: Dev Note
ls_header-tdobject  = 'ZBUG'.          " Text Object
ls_header-tdspras   = sy-langu.        " Language

CALL FUNCTION 'READ_TEXT'
  EXPORTING
    id        = ls_header-tdid
    language  = ls_header-tdspras
    name      = ls_header-tdname
    object    = ls_header-tdobject
  TABLES
    lines     = lt_lines
  EXCEPTIONS
    not_found = 1
    OTHERS    = 2.

" --- Ghi Long Text ---
CALL FUNCTION 'SAVE_TEXT'
  EXPORTING
    header   = ls_header
    insert   = 'X'           " 'X' = tạo mới nếu chưa có
  TABLES
    lines    = lt_lines
  EXCEPTIONS
    OTHERS   = 1.
COMMIT WORK.
```

✅ **Checkpoint: SE75** → Text Object `ZBUG` → 3 Text IDs (`Z001`, `Z002`, `Z003`) active.

Bước A8: Status Code Migration Script

Mục tiêu: Migrate status codes cũ sang 9-state model mới. Đây là bước **CRITICAL** — phải chạy SAU khi A4 hoàn thành (IS_DEL field tồn tại) và TRƯỚC khi Phase B code goes live.

⚠️ CONFLICT GIỮA HỆ THỐNG CŨ VÀ MỚI:

Status Code	Ý nghĩa CŨ (developer-guide.md)	Ý nghĩa MỚI (9-state)
4	Fixed	**Pending**
5	Closed	**Fixed**
6	Deleted	**Resolved**
7	*(không có)*	**Closed**
R	*(không có)*	**Rejected**

Nếu KHÔNG migrate, bugs cũ có status=4 (Fixed) sẽ bị hiểu nhầm là Pending!

Tạo report `Z_BUG_MIGRATE_STATUS` trong **SE38** → chạy 1 lần duy nhất:

```
REPORT z_bug_migrate_status.

*&---------------------------------------------------------------------*
*& Status Code Migration: Old → New 9-State Model
*& Chạy 1 lần SAU Phase A4 (IS_DEL field exists), TRƯỚC Phase B
*&---------------------------------------------------------------------*

DATA: lv_cnt_4  TYPE i,  " Old Fixed
      lv_cnt_5  TYPE i,  " Old Closed
      lv_cnt_6  TYPE i,  " Old Deleted
      lv_answer TYPE char1.

" ====== PRE-CHECK: Đếm records cần migrate ======
SELECT COUNT(*) FROM zbug_tracker INTO @lv_cnt_4 WHERE status = '4'.
SELECT COUNT(*) FROM zbug_tracker INTO @lv_cnt_5 WHERE status = '5'.
SELECT COUNT(*) FROM zbug_tracker INTO @lv_cnt_6 WHERE status = '6'.

DATA: lv_total TYPE i.
lv_total = lv_cnt_4 + lv_cnt_5 + lv_cnt_6.

WRITE: / '=== STATUS MIGRATION PRE-CHECK ==='.
WRITE: / 'Status 4 (Old Fixed → New 5 Fixed):', lv_cnt_4, 'records'.
WRITE: / 'Status 5 (Old Closed → New 7 Closed):', lv_cnt_5, 'records'.
```

```
WRITE: / 'Status 6 (Old Deleted → IS_DEL=X):', lv_cnt_6, 'records'.
WRITE: / 'Total records to migrate:', lv_total.
SKIP.

IF lv_total = 0.
  WRITE: / 'Không có records nào cần migrate. Kết thúc.'.
  RETURN.
ENDIF.

" ====== CONFIRMATION ======
CALL FUNCTION 'POPUP_TO_CONFIRM'
  EXPORTING
    titlebar      = 'Status Migration'
    text_question = |Migrate { lv_total } records? This cannot be undone.|
    text_button_1 = 'Yes, Migrate'
    text_button_2 = 'Cancel'
  IMPORTING
    answer        = lv_answer.

IF lv_answer <> '1'.
  WRITE: / 'Migration cancelled by user.'.
  RETURN.
ENDIF.

" ====== STEP 1: Remap status codes ======
" THỨ TỰ QUAN TRỌNG: Phải remap 5→7 TRƯỚC, rồi mới 4→5
" Nếu làm ngược (4→5 trước), thì khi remap 5→7 sẽ bắt cả records vừa đổi!

" Step 1a: Old 5 (Closed) → New 7 (Closed)
UPDATE zbug_tracker SET status = '7',
                        aenam  = @sy-uname,
                        aedat  = @sy-datum,
                        aezet  = @sy-uzeit
  WHERE status = '5'.
DATA(lv_migrated_5) = sy-dbcnt.
WRITE: / 'Step 1a: Status 5→7 (Closed):', lv_migrated_5, 'records updated'.

" Step 1b: Old 4 (Fixed) → New 5 (Fixed)
UPDATE zbug_tracker SET status = '5',
                        aenam  = @sy-uname,
                        aedat  = @sy-datum,
                        aezet  = @sy-uzeit
  WHERE status = '4'.
DATA(lv_migrated_4) = sy-dbcnt.
WRITE: / 'Step 1b: Status 4→5 (Fixed):', lv_migrated_4, 'records updated'.

" ====== STEP 2: Convert old Deleted (status=6) → IS_DEL='X' ======
" Set status to '7' (Closed) vì bug đã bị xóa = effectively closed
UPDATE zbug_tracker SET is_del = 'X',
```

```
                        status = '7',
                        aenam  = @sy-uname,
                        aedat  = @sy-datum,
                        aezet  = @sy-uzeit
  WHERE status = '6'.
DATA(lv_migrated_6) = sy-dbcnt.
WRITE: / 'Step 2: Status 6→IS_DEL=X:', lv_migrated_6, 'records soft-deleted'.

" ====== STEP 3: Update ZBUG_HISTORY references ======
" Remap old_value/new_value trong history (cùng thứ tự: 5→7 trước, 4→5 sau)
UPDATE zbug_history SET new_value = '7' WHERE new_value = '5' AND action_type =
'ST'.
UPDATE zbug_history SET old_value = '7' WHERE old_value = '5' AND action_type =
'ST'.
UPDATE zbug_history SET new_value = '5' WHERE new_value = '4' AND action_type =
'ST'.
UPDATE zbug_history SET old_value = '5' WHERE old_value = '4' AND action_type =
'ST'.
WRITE: / 'Step 3: ZBUG_HISTORY old/new values remapped'.

" ====== STEP 4: COMMIT ======
COMMIT WORK AND WAIT.
WRITE: / ''.
WRITE: / '=== MIGRATION COMPLETE ==='.
lv_total = lv_migrated_4 + lv_migrated_5 + lv_migrated_6.
WRITE: / 'Total migrated:', lv_total, 'records'.

" ====== POST-CHECK ======
SKIP.
WRITE: / '=== POST-MIGRATION VERIFICATION ==='.
SELECT COUNT(*) FROM zbug_tracker INTO @DATA(lv_check_4) WHERE status = '4'.
SELECT COUNT(*) FROM zbug_tracker INTO @DATA(lv_check_6)
  WHERE status = '6' AND is_del <> 'X'.
WRITE: / 'Records still at status 4 (should be 0 unless new Pending):',
lv_check_4.
WRITE: / 'Records at status 6 without IS_DEL (should be 0):', lv_check_6.

IF lv_check_4 = 0 AND lv_check_6 = 0.
  WRITE: / '✅ Migration verified successfully!'.
ELSE.
  WRITE: / '⚠ WARNING: Unexpected records found. Please check manually.'.
ENDIF.
```

⚠️ Chạy report này ĐÚNG 1 LẦN, sau A4 và TRƯỚC Phase B.

Nếu chạy nhầm 2 lần, Step 1 sẽ không gây hại (vì không còn records status=4/5 cũ), nhưng cần verify lại.

✅ **Checkpoint sau khi chạy:** - **SE16** → `ZBUG_TRACKER` → không còn status='6' với IS_DEL='' (space) - **SE16** → `ZBUG_TRACKER` → tất cả bug cũ "Fixed" giờ có status='5', "Closed" có status='7' - Report output hiện "Migration verified successfully!"

TỔNG KẾT PHASE A

Sau khi hoàn thành Phase A, bạn phải có:

- ☒ **4 Domains mới:** `ZDOM_PROJECT_ID`, `ZDOM_PRJ_STATUS`, `ZDOM_SEVERITY`, `ZDOM_IS_DEL`
- ☒ **6 Data Elements mới:** `ZDE_PROJECT_ID`, `ZDE_PRJ_NAME`, `ZDE_PRJ_DESC`, `ZDE_PRJ_STATUS`, `ZDE_SEVERITY`, `ZDE_IS_DEL`
- ☒ **2 Bảng mới:** `ZBUG_PROJECT` (16 fields), `ZBUG_USER_PROJEC` (9 fields)
- ☒ **2 Bảng cập nhật:** `ZBUG_TRACKER` (+13 fields: PROJECT_ID, SEVERITY, VERIFY_TESTER_ID, APPROVED_BY/AT, CLOSED_AT, ERNAM/DAT/ZET, AENAM/DAT/ZET, IS_DEL), `ZBUG_USERS` (+4 fields)
- ☒ **1 Message Class:** `ZBUG_MSG` (33+ messages, EN + VI)
- ☒ **1 Text Object:** `ZBUG` (3 Text IDs: `Z001`, `Z002`, `Z003`)
- ☒ **1 Migration Report:** `Z_BUG_MIGRATE_STATUS` (đã chạy thành công)

Kiểm tra cuối cùng:

1. SE11 → `ZBUG_PROJECT` → Active
2. SE11 → `ZBUG_USER_PROJEC` → Active
3. SE11 → `ZBUG_TRACKER` → có `PROJECT_ID`, `SEVERITY`, `ERNAM`, `ERDAT`, `ERZET`, `IS_DEL`
4. SE11 → `ZBUG_USERS` → có `AENAM`, `IS_DEL`, `EMAIL` obligatory
5. SE91 → `ZBUG_MSG` → 33+ messages
6. SE75 → `ZBUG` → 3 Text IDs
7. `Z_BUG_MIGRATE_STATUS` → đã chạy, output hiện "Migration verified successfully!"

👉 **Chuyển sang Phase B: Business Logic Update** # HƯỚNG DẪN TRIỂN KHAI CHI TIẾT — PHASE B: BUSINESS LOGIC UPDATE

Dự án: SAP Bug Tracking Management System **Ngày:** 24/03/2026 | **Phiên bản:** 5.0 (Module Pool Integration) **Thời gian ước tính:** 2 ngày (25-27/03) **Yêu cầu:** Hoàn thành Phase A trước khi bắt đầu Phase B

⚠️ **QUAN TRỌNG:** Phải chạy report `Z_BUG_MIGRATE_STATUS` (Phase A, Bước A8) trước khi activate bất kỳ FM nào trong Phase B. Nếu không, status codes cũ (4=Fixed, 5=Closed) sẽ conflict với logic mới (4=Pending, 5=Fixed).

Bước B1: Mở rộng `Z_BUG_CHECK_PERMISSION`

Mục tiêu: Thêm Project permissions + user-project membership check.

Vào **SE37** → mở FM `Z_BUG_CHECK_PERMISSION` → **Change**.

Thêm IMPORTING parameter mới:

```
IV_PROJECT_ID TYPE ZDE_PROJECT_ID OPTIONAL
```

Source code cập nhật (thay thế toàn bộ):

```
FUNCTION z_bug_check_permission.
*"----------------------------------------------------------------------
*"*"Local Interface:
*"  IMPORTING
*"     VALUE(IV_USER) TYPE  ZDE_USERNAME
*"     VALUE(IV_BUG_ID) TYPE  ZDE_BUG_ID OPTIONAL
*"     VALUE(IV_ACTION) TYPE  CHAR20
*"     VALUE(IV_PROJECT_ID) TYPE  ZDE_PROJECT_ID OPTIONAL
*"  EXPORTING
*"     VALUE(EV_ALLOWED) TYPE  CHAR1
*"     VALUE(EV_MESSAGE) TYPE  STRING
*"----------------------------------------------------------------------

  DATA: ls_user    TYPE zbug_users,
        ls_bug     TYPE zbug_tracker,
        ls_project TYPE zbug_project,
        lv_count   TYPE i.

  " 1. Lấy thông tin User
  SELECT SINGLE * FROM zbug_users INTO @ls_user
    WHERE user_id = @iv_user AND is_del <> 'X'.

  IF sy-subrc <> 0.
    ev_allowed = 'N'.
    MESSAGE s029(zbug_msg) WITH iv_user INTO ev_message.
    RETURN.
  ENDIF.

  " 2. ĐẶC QUYỀN: Manager (M) có toàn quyền
  IF ls_user-role = 'M'.
    ev_allowed = 'Y'.
    RETURN.
  ENDIF.

  " 3. Lấy thông tin Bug (nếu có)
  IF iv_bug_id IS NOT INITIAL.
    SELECT SINGLE * FROM zbug_tracker INTO @ls_bug
      WHERE bug_id = @iv_bug_id AND is_del <> 'X'.
  ENDIF.

  " 4. CHECK USER-PROJECT MEMBERSHIP (nếu có project context)
  DATA(lv_prj) = COND #( WHEN iv_project_id IS NOT INITIAL THEN iv_project_id
                         WHEN ls_bug-project_id IS NOT INITIAL THEN ls_bug-
```

```
project_id
                          ELSE '' ).
  IF lv_prj IS NOT INITIAL.
    SELECT COUNT(*) FROM zbug_user_project INTO @lv_count
      WHERE user_id = @iv_user AND project_id = @lv_prj.
    IF lv_count = 0.
      ev_allowed = 'N'.
      MESSAGE s004(zbug_msg) INTO ev_message.
      RETURN.
    ENDIF.
  ENDIF.

  " 5. RẼ NHÁNH KIỂM TRA QUYỀN THEO HÀNH ĐỘNG
  CASE iv_action.

    " === BUG ACTIONS ===
    WHEN 'CREATE'.
      IF ls_user-role = 'T'.
        ev_allowed = 'Y'.
      ELSE.
        ev_allowed = 'N'.
        MESSAGE s005(zbug_msg) INTO ev_message.
      ENDIF.

    WHEN 'UPDATE_STATUS'.
      " Config Bug: Tester tự xử lý
      IF ls_bug-bug_type = 'F' AND ls_user-role = 'T' AND ls_bug-dev_id =
iv_user.
        ev_allowed = 'Y'.
      " Developer update bug do mình giữ
      ELSEIF ls_user-role = 'D' AND ls_bug-dev_id = iv_user.
        ev_allowed = 'Y'.
      " Tester update khi Bug mới tạo (Status 1)
      ELSEIF ls_user-role = 'T' AND ls_bug-status = '1'.
        ev_allowed = 'Y'.
      ELSE.
        ev_allowed = 'N'.
        MESSAGE s006(zbug_msg) INTO ev_message.
      ENDIF.

    WHEN 'DELETE_BUG'.
      ev_allowed = 'N'.
      MESSAGE s006(zbug_msg) INTO ev_message.

    WHEN 'UPLOAD_REPORT'.
      IF ls_user-role = 'T' AND ls_bug-tester_id = iv_user.
        ev_allowed = 'Y'.
      ELSE.
        ev_allowed = 'N'.
```

```
        MESSAGE s006(zbug_msg) INTO ev_message.
      ENDIF.

    WHEN 'UPLOAD_FIX'.
      IF ls_bug-bug_type = 'F' AND ls_user-role = 'T' AND ls_bug-dev_id =
iv_user.
        ev_allowed = 'Y'.
      ELSEIF ls_user-role = 'D' AND ls_bug-dev_id = iv_user.
        ev_allowed = 'Y'.
      ELSE.
        ev_allowed = 'N'.
        MESSAGE s006(zbug_msg) INTO ev_message.
      ENDIF.

    WHEN 'UPLOAD_VERIFY'.
      IF ls_user-role = 'T'.
        ev_allowed = 'Y'.
      ELSE.
        ev_allowed = 'N'.
        MESSAGE s006(zbug_msg) INTO ev_message.
      ENDIF.

    " === PROJECT ACTIONS ===
    WHEN 'CREATE_PROJECT'.
      ev_allowed = 'N'.
      MESSAGE s007(zbug_msg) INTO ev_message.

    WHEN 'CHANGE_PROJECT'.
      ev_allowed = 'N'.
      MESSAGE s007(zbug_msg) INTO ev_message.

    WHEN 'DELETE_PROJECT'.
      ev_allowed = 'N'.
      MESSAGE s007(zbug_msg) INTO ev_message.

    WHEN 'VIEW_PROJECT'.
      ev_allowed = 'Y'.

    WHEN 'ADD_USER_PROJECT'.
      ev_allowed = 'N'.
      MESSAGE s007(zbug_msg) INTO ev_message.

    WHEN OTHERS.
      ev_allowed = 'N'.
      ev_message = 'Unknown action'.
  ENDCASE.

ENDFUNCTION.
```

Nhấn **Activate**.

✅ **Checkpoint: SE37** → `Z_BUG_CHECK_PERMISSION` → Test:

- `IV_USER` = (tester), `IV_ACTION` = `CREATE_PROJECT` → `EV_ALLOWED` = `N`
- `IV_USER` = (manager), `IV_ACTION` = `CREATE_PROJECT` → `EV_ALLOWED` = `Y`
- `IV_USER` = (tester), `IV_ACTION` = `CREATE`, `IV_PROJECT_ID` = `PRJ001` → check membership

Bước B2: Cập nhật `Z_BUG_CREATE`

Mục tiêu: Thêm `PROJECT_ID`, `SEVERITY`, user validation, project validation.

Vào **SE37** → mở FM `Z_BUG_CREATE` → **Change**.

Thêm IMPORTING parameters:

```
IV_PROJECT_ID TYPE ZDE_PROJECT_ID
IV_SEVERITY   TYPE ZDE_SEVERITY DEFAULT '4'
```

Source code cập nhật:

```
FUNCTION z_bug_create.
*"----------------------------------------------------------------------
*"*"Local Interface:
*"  IMPORTING
*"     VALUE(IV_TITLE) TYPE  ZDE_BUG_TITLE
*"     VALUE(IV_DESC) TYPE  ZDE_BUG_DESC
*"     VALUE(IV_MODULE) TYPE  ZDE_SAP_MODULE
*"     VALUE(IV_PRIORITY) TYPE  ZDE_PRIORITY DEFAULT 'M'
*"     VALUE(IV_DEV_ID) TYPE  ZDE_USERNAME OPTIONAL
*"     VALUE(IV_BUG_TYPE) TYPE  ZDE_BUG_TYPE OPTIONAL
*"     VALUE(IV_ATT_PATH) TYPE  ZDE_BUG_ATT_PATH OPTIONAL
*"     VALUE(IV_PROJECT_ID) TYPE  ZDE_PROJECT_ID
*"     VALUE(IV_SEVERITY) TYPE  ZDE_SEVERITY DEFAULT '4'
*"  EXPORTING
*"     VALUE(EV_BUG_ID) TYPE  ZDE_BUG_ID
*"     VALUE(EV_SUCCESS) TYPE  CHAR1
*"     VALUE(EV_MESSAGE) TYPE  STRING
*"----------------------------------------------------------------------

  DATA: ls_bug     TYPE zbug_tracker,
        lv_number  TYPE numc10,
        lv_count   TYPE i.

  " ====== VALIDATION BLOCK ======

  " 1. Validate user exists + active
  SELECT COUNT(*) FROM zbug_users INTO @lv_count
    WHERE user_id = @sy-uname AND is_del <> 'X'.
```

```
IF lv_count = 0.
  ev_success = 'N'.
  MESSAGE s029(zbug_msg) WITH sy-uname INTO ev_message.
  RETURN.
ENDIF.

" 2. Validate project exists + InProcess
SELECT SINGLE project_status FROM zbug_project INTO @DATA(lv_prj_status)
  WHERE project_id = @iv_project_id AND is_del <> 'X'.
IF sy-subrc <> 0.
  ev_success = 'N'.
  MESSAGE s021(zbug_msg) WITH iv_project_id INTO ev_message.
  RETURN.
ENDIF.
IF lv_prj_status <> '2'. " Must be InProcess
  ev_success = 'N'.
  MESSAGE s022(zbug_msg) INTO ev_message.
  RETURN.
ENDIF.

" 3. Validate user belongs to project
SELECT COUNT(*) FROM zbug_user_project INTO @lv_count
  WHERE user_id = @sy-uname AND project_id = @iv_project_id.
IF lv_count = 0.
  ev_success = 'N'.
  MESSAGE s004(zbug_msg) INTO ev_message.
  RETURN.
ENDIF.

" 4. Severity + Bug Type cross validation
IF iv_bug_type IS INITIAL.
  ls_bug-bug_type = 'C'. " Default Code bug
ELSE.
  ls_bug-bug_type = iv_bug_type.
ENDIF.

" Dump/VeryHigh/High severity Code bugs → force High priority
IF ( iv_severity = '1' OR iv_severity = '2' OR iv_severity = '3' )
   AND ls_bug-bug_type = 'C'.
  ls_bug-priority = 'H'.
ELSE.
  ls_bug-priority = iv_priority.
ENDIF.

" ====== BUG ID GENERATION ======

CALL FUNCTION 'NUMBER_GET_NEXT'
  EXPORTING
    nr_range_nr             = '01'
```

```
    object                  = 'ZNRO_BUG'
  IMPORTING
    number                  = lv_number
  EXCEPTIONS
    interval_not_found      = 1
    number_range_not_intern = 2
    object_not_found        = 3
    quantity_is_0           = 4
    quantity_is_not_1       = 5
    interval_overflow       = 6
    buffer_overflow         = 7
    OTHERS                  = 8.

IF sy-subrc <> 0.
  ev_success = 'N'.
  MESSAGE s011(zbug_msg) INTO ev_message.
  RETURN.
ENDIF.

CONCATENATE 'BUG' lv_number+3(7) INTO ev_bug_id.

" ====== FILL BUG RECORD ======

ls_bug-bug_id       = ev_bug_id.
ls_bug-title        = iv_title.
ls_bug-desc_text    = iv_desc.
ls_bug-sap_module   = iv_module.
ls_bug-tester_id    = sy-uname.
ls_bug-created_at   = sy-datum.
ls_bug-created_time = sy-uzeit.
ls_bug-ernam        = sy-uname.
ls_bug-erdat        = sy-datum.
ls_bug-erzet        = sy-uzeit.
ls_bug-project_id   = iv_project_id.
ls_bug-severity     = iv_severity.

" Evidence
IF iv_att_path IS NOT INITIAL.
  ls_bug-att_report = iv_att_path.
ENDIF.

" ====== WORKFLOW BRANCHING ======

IF ls_bug-bug_type = 'F'.
  " Config Bug: Tester tự xử lý
  ls_bug-status = '2'.
  ls_bug-dev_id = sy-uname.
ELSE.
  " Code Bug: New, chờ Auto-Assign
```

```
    ls_bug-status = '1'.
    ls_bug-dev_id = ''.
  ENDIF.

  " ====== DATABASE INSERT ======
  INSERT zbug_tracker FROM ls_bug.

  IF sy-subrc = 0.
    ev_success = 'Y'.
    MESSAGE s001(zbug_msg) WITH ev_bug_id INTO ev_message.

    " Log history
    CALL FUNCTION 'Z_BUG_LOG_HISTORY'
      EXPORTING
        iv_bug_id      = ev_bug_id
        iv_action_type = 'CR'
        iv_old_value   = ''
        iv_new_value   = iv_title
        iv_reason      = 'New bug created'.

    " Send email notification
    CALL FUNCTION 'Z_BUG_SEND_EMAIL'
      EXPORTING
        iv_bug_id = ev_bug_id
        iv_event  = 'CREATE'.

    COMMIT WORK AND WAIT.
  ELSE.
    ev_success = 'N'.
    MESSAGE s010(zbug_msg) INTO ev_message.
    ROLLBACK WORK.
  ENDIF.

ENDFUNCTION.
```

Nhấn **Activate**.

✅ **Checkpoint: SE37** → `Z_BUG_CREATE` → Test:

- Truyền `IV_PROJECT_ID` không tồn tại → "Project does not exist"
- Truyền phần tử Valid → Bug tạo thành công với `PROJECT_ID` + `SEVERITY`

Bước B3: Chuẩn hóa Status States — `Z_BUG_UPDATE_STATUS` (Full Rewrite)

Mục tiêu: Refactor `Z_BUG_UPDATE_STATUS` cho 9 status states mới với full validation.

Vào **SE37** → `Z_BUG_UPDATE_STATUS` → **Change**.

Bảng Transition hợp lệ:

From	→ Valid To
`1`	`2` (assign), `W` (no dev)
`W`	`2` (manager assign)
`2`	`3` (dev start), `R` (reject)
`3`	`5` (fixed), `4` (pending)
`4`	`3` (resume)
`5`	`6` (verify pass), `3` (fail)
`6`	`7` (close)
`7`	*(terminal — no transition)*
`R`	`2` (reassign)

Source code cập nhật (thay thế toàn bộ):

```
FUNCTION z_bug_update_status.
*"----------------------------------------------------------------------
*"*"Local Interface:
*"  IMPORTING
*"     VALUE(IV_BUG_ID) TYPE  ZDE_BUG_ID
*"     VALUE(IV_NEW_STATUS) TYPE  CHAR1
*"     VALUE(IV_DEV_ID) TYPE  ZDE_USERNAME OPTIONAL
*"     VALUE(IV_REASON) TYPE  STRING OPTIONAL
*"  EXPORTING
*"     VALUE(EV_SUCCESS) TYPE  CHAR1
*"     VALUE(EV_MESSAGE) TYPE  STRING
*"----------------------------------------------------------------------

  DATA: ls_bug   TYPE zbug_tracker,
        lv_valid TYPE abap_bool VALUE abap_false.

  " ====== 1. READ CURRENT BUG ======
  SELECT SINGLE * FROM zbug_tracker INTO @ls_bug
    WHERE bug_id = @iv_bug_id AND is_del <> 'X'.

  IF sy-subrc <> 0.
    ev_success = 'N'.
    MESSAGE s029(zbug_msg) WITH iv_bug_id INTO ev_message.
    RETURN.
  ENDIF.

  " ====== 2. CHECK BUG NOT CLOSED ======
```

```
IF ls_bug-status = '7'.
  ev_success = 'N'.
  MESSAGE s033(zbug_msg) INTO ev_message.
  RETURN.
ENDIF.

" ====== 3. PERMISSION CHECK ======
CALL FUNCTION 'Z_BUG_CHECK_PERMISSION'
  EXPORTING
    iv_user   = sy-uname
    iv_bug_id = iv_bug_id
    iv_action = 'UPDATE_STATUS'
  IMPORTING
    ev_allowed = DATA(lv_allowed)
    ev_message = ev_message.

IF lv_allowed <> 'Y'.
  ev_success = 'N'.
  RETURN.
ENDIF.

" ====== 4. TRANSITION VALIDATION ======
CASE ls_bug-status.
  WHEN '1'.  " New
    IF iv_new_status = '2' OR iv_new_status = 'W'.
      lv_valid = abap_true.
    ENDIF.
  WHEN 'W'.  " Waiting
    IF iv_new_status = '2'.
      lv_valid = abap_true.
    ENDIF.
  WHEN '2'.  " Assigned
    IF iv_new_status = '3' OR iv_new_status = 'R'.
      lv_valid = abap_true.
    ENDIF.
  WHEN '3'.  " In Progress
    IF iv_new_status = '5' OR iv_new_status = '4'.
      lv_valid = abap_true.
    ENDIF.
  WHEN '4'.  " Pending
    IF iv_new_status = '3'.
      lv_valid = abap_true.
    ENDIF.
  WHEN '5'.  " Fixed
    IF iv_new_status = '6' OR iv_new_status = '3'.
      lv_valid = abap_true.
    ENDIF.
  WHEN '6'.  " Resolved
    IF iv_new_status = '7'.
```

```
      lv_valid = abap_true.
    ENDIF.
  WHEN '7'.  " Closed — terminal
    lv_valid = abap_false.
  WHEN 'R'.  " Rejected
    IF iv_new_status = '2'.
      lv_valid = abap_true.
    ENDIF.
ENDCASE.

IF lv_valid = abap_false.
  ev_success = 'N'.
  DATA: lv_old_text TYPE char20, lv_new_text TYPE char20.
  lv_old_text = SWITCH #( ls_bug-status
    WHEN '1' THEN 'New'        WHEN 'W' THEN 'Waiting'
    WHEN '2' THEN 'Assigned'   WHEN '3' THEN 'InProgress'
    WHEN '4' THEN 'Pending'    WHEN '5' THEN 'Fixed'
    WHEN '6' THEN 'Resolved'   WHEN '7' THEN 'Closed'
    WHEN 'R' THEN 'Rejected' ).
  lv_new_text = SWITCH #( iv_new_status
    WHEN '1' THEN 'New'        WHEN 'W' THEN 'Waiting'
    WHEN '2' THEN 'Assigned'   WHEN '3' THEN 'InProgress'
    WHEN '4' THEN 'Pending'    WHEN '5' THEN 'Fixed'
    WHEN '6' THEN 'Resolved'   WHEN '7' THEN 'Closed'
    WHEN 'R' THEN 'Rejected' ).
  MESSAGE s018(zbug_msg) WITH lv_old_text lv_new_text INTO ev_message.
  RETURN.
ENDIF.

" ====== 5. EVIDENCE VALIDATION ======
" Chuyển sang Fixed (5): phải có TESTCASE file
IF iv_new_status = '5'.
  IF ls_bug-att_fix IS INITIAL.
    " Check GOS
    DATA: lt_files TYPE sbdst_files.
    CALL FUNCTION 'Z_BUG_GOS_LIST'
      EXPORTING iv_bug_id = iv_bug_id
      IMPORTING et_files  = lt_files.
    " Tìm file TESTCASE trong GOS
    DATA(lv_has_testcase) = abap_false.
    LOOP AT lt_files TRANSPORTING NO FIELDS
      WHERE comp_id CS 'TESTCASE' OR comp_id CS 'testcase'.
      lv_has_testcase = abap_true. EXIT.
    ENDLOOP.
    IF lv_has_testcase = abap_false.
      ev_success = 'N'.
      MESSAGE s016(zbug_msg) INTO ev_message.
      RETURN.
    ENDIF.
```

```
  ENDIF.
ENDIF.

" Chuyển sang Resolved (6): phải có CONFIRM file
IF iv_new_status = '6'.
  IF ls_bug-att_verify IS INITIAL.
    CALL FUNCTION 'Z_BUG_GOS_LIST'
      EXPORTING iv_bug_id = iv_bug_id
      IMPORTING et_files  = lt_files.
    DATA(lv_has_confirm) = abap_false.
    LOOP AT lt_files TRANSPORTING NO FIELDS
      WHERE comp_id CS 'CONFIRM' OR comp_id CS 'confirm'.
      lv_has_confirm = abap_true. EXIT.
    ENDLOOP.
    IF lv_has_confirm = abap_false.
      ev_success = 'N'.
      MESSAGE s017(zbug_msg) INTO ev_message.
      RETURN.
    ENDIF.
  ENDIF.
ENDIF.

" ====== 6. SPECIAL FIELD UPDATES ======
DATA(lv_old_status) = ls_bug-status.
ls_bug-status = iv_new_status.

" →2 (Assigned): Set DEV_ID
IF iv_new_status = '2' AND iv_dev_id IS NOT INITIAL.
  ls_bug-dev_id = iv_dev_id.
ENDIF.

" →6 (Resolved): Tester đã verify
IF iv_new_status = '6'.
  ls_bug-verify_tester_id = sy-uname.
ENDIF.

" →7 (Closed): Manager close
IF iv_new_status = '7'.
  ls_bug-closed_at   = sy-datum.
  ls_bug-approved_by  = sy-uname.
  ls_bug-approved_at  = sy-datum.
ENDIF.

" ====== 7. AUDIT FIELDS ======
ls_bug-aenam = sy-uname.
ls_bug-aedat = sy-datum.
ls_bug-aezet = sy-uzeit.

" ====== 8. DATABASE UPDATE ======
```

```
  UPDATE zbug_tracker FROM ls_bug.

  IF sy-subrc = 0.
    ev_success = 'Y'.
    MESSAGE s002(zbug_msg) WITH iv_bug_id INTO ev_message.

    " Log history
    CALL FUNCTION 'Z_BUG_LOG_HISTORY'
      EXPORTING
        iv_bug_id      = iv_bug_id
        iv_action_type = 'ST'
        iv_old_value   = lv_old_status
        iv_new_value   = iv_new_status
        iv_reason      = iv_reason.

    " Send email
    CALL FUNCTION 'Z_BUG_SEND_EMAIL'
      EXPORTING
        iv_bug_id = iv_bug_id
        iv_event  = 'STATUS_CHANGE'.

    COMMIT WORK AND WAIT.
  ELSE.
    ev_success = 'N'.
    MESSAGE s010(zbug_msg) INTO ev_message.
    ROLLBACK WORK.
  ENDIF.

ENDFUNCTION.
```

Nhấn **Activate**.

✅ **Checkpoint: SE37** → `Z_BUG_UPDATE_STATUS` → Test:

- Status `1` → `5` (Fixed) → FAIL "Invalid transition from New to Fixed"
- Status `1` → `2` (Assigned) → PASS
- Status `3` → `5` (Fixed) without TESTCASE file → FAIL "Upload TESTCASE file…"
- Status `6` → `7` (Closed) → PASS, CLOSED_AT + APPROVED_BY filled

---

Bước B4: GOS File Storage Integration

Mục tiêu: Tạo cơ chế upload file qua GOS thay local path.

Dùng BDS API:

Tạo FM mới `Z_BUG_GOS_UPLOAD` trong Function Group `ZBUG_FG`:

```
FUNCTION z_bug_gos_upload.
*"----------------------------------------------------------------------
*"*"Local Interface:
*"  IMPORTING
*"     VALUE(IV_BUG_ID) TYPE  ZDE_BUG_ID
*"     VALUE(IV_FILE_PATH) TYPE  STRING
*"     VALUE(IV_FILE_TYPE) TYPE  CHAR20
*"  EXPORTING
*"     VALUE(EV_SUCCESS) TYPE  CHAR1
*"     VALUE(EV_MESSAGE) TYPE  STRING
*"----------------------------------------------------------------------

  DATA: lt_file_content TYPE TABLE OF soli,
        lv_file_size    TYPE i,
        lv_file_name    TYPE string.

  " 1. Đọc file từ frontend
  CALL FUNCTION 'GUI_UPLOAD'
    EXPORTING
      filename   = iv_file_path
      filetype   = 'BIN'
    IMPORTING
      filelength = lv_file_size
    TABLES
      data_tab   = lt_file_content
    EXCEPTIONS
      OTHERS     = 1.

  IF sy-subrc <> 0.
    ev_success = 'N'.
    MESSAGE s000(zbug_msg) WITH 'Failed to read' 'file from local' INTO
ev_message.
    RETURN.
  ENDIF.

  " 2. Lấy tên file (basename)
  DATA: lt_parts TYPE TABLE OF string.
  SPLIT iv_file_path AT '\' INTO TABLE lt_parts.
  READ TABLE lt_parts INTO lv_file_name INDEX lines( lt_parts ).

  " 3. Lưu vào BDS
  DATA: lo_bds       TYPE REF TO cl_bds_document_set,
        lt_files     TYPE sbdst_files,
        ls_file      TYPE sbdst_file,
        lt_signature TYPE sbdst_signature,
        ls_signature TYPE sbdst_signat.

  CREATE OBJECT lo_bds.
```

```
  ls_signature-prop_name  = 'DESCRIPTION'.
  ls_signature-prop_value = |{ iv_file_type }: { lv_file_name }|.
  APPEND ls_signature TO lt_signature.

  ls_file-doc_count = 1.
  ls_file-comp_count = 1.
  ls_file-comp_id = lv_file_name.
  ls_file-mimetype = 'application/octet-stream'.
  ls_file-comp_size = lv_file_size.
  APPEND ls_file TO lt_files.

  TRY.
    lo_bds->create_with_table(
      EXPORTING classname  = 'ZBUG_EVIDENCE'
                classtype  = 'OT'
                object_key = CONV #( iv_bug_id )
      CHANGING  components = lt_files
                content    = lt_file_content
                signature  = lt_signature ).
    COMMIT WORK.
    ev_success = 'Y'.
    ev_message = |File { lv_file_name } uploaded successfully|.
  CATCH cx_bds_kernel INTO DATA(lx_error).
    ev_success = 'N'.
    ev_message = lx_error->get_text( ).
    ROLLBACK WORK.
  ENDTRY.

ENDFUNCTION.
```

Tạo FM Z_BUG_GOS_LIST:

```
FUNCTION z_bug_gos_list.
*"----------------------------------------------------------------------
*"*"Local Interface:
*"  IMPORTING
*"     VALUE(IV_BUG_ID) TYPE  ZDE_BUG_ID
*"  EXPORTING
*"     VALUE(ET_FILES) TYPE  SBDST_FILES
*"----------------------------------------------------------------------

  DATA: lo_bds       TYPE REF TO cl_bds_document_set,
        lt_signature TYPE sbdst_signature.

  CREATE OBJECT lo_bds.

  TRY.
    lo_bds->get_info(
      EXPORTING classname  = 'ZBUG_EVIDENCE'
                classtype  = 'OT'
```

```
                object_key = CONV #( iv_bug_id )
      CHANGING  components = et_files
                signature  = lt_signature ).
  CATCH cx_bds_kernel.
    CLEAR et_files.
  ENDTRY.

ENDFUNCTION.
```

Nhấn **Activate** cả 2 FMs.

Bước B5: SmartForm Email Template

Mục tiêu: Tạo SmartForm `ZBUG_EMAIL_FORM` cho email notification.

Vào T-code **SMARTFORMS** → nhập `ZBUG_EMAIL_FORM` → **Create**.

Step 1: Form Interface

- IMPORTING:
  - `IV_BUG_ID TYPE ZDE_BUG_ID`
  - `IV_EVENT  TYPE CHAR20`
  - `IS_BUG    TYPE ZBUG_TRACKER`
  - `IV_EVENT_TEXT TYPE STRING`

Step 2: Global Definitions

- Global Data:
  - `GV_TITLE TYPE STRING`
  - `GV_BODY  TYPE STRING`

Step 3: Form Layout

- Pages and Windows → MAIN Window → Main Window
- Text Element **BUG_INFO:**

```
Bug ID:    &IS_BUG-BUG_ID&
Title:     &IS_BUG-TITLE&
Project:   &IS_BUG-PROJECT_ID&
Status:    &IS_BUG-STATUS&
```

Nhấn **Save** → **Activate**.

Bước B6: Cập nhật `Z_BUG_SEND_EMAIL`

Mục tiêu: Sửa FM email để dùng SmartForm + event-based recipients.

```
FUNCTION z_bug_send_email.
*"----------------------------------------------------------------------
*"*"Local Interface:
```

```
*"  IMPORTING
*"     VALUE(IV_BUG_ID) TYPE  ZDE_BUG_ID
*"     VALUE(IV_EVENT) TYPE  CHAR20
*"----------------------------------------------------------------------

  DATA: ls_bug       TYPE zbug_tracker,
        lt_recip     TYPE TABLE OF zbug_users,
        lo_send_req  TYPE REF TO cl_bcs,
        lo_doc       TYPE REF TO cl_document_bcs,
        lo_sender    TYPE REF TO cl_sapuser_bcs,
        lo_recip     TYPE REF TO if_recipient_bcs,
        lv_subject   TYPE so_obj_des,
        lt_body      TYPE bcsy_text,
        ls_body      TYPE soli,
        lv_sent      TYPE os_boolean.

  " 1. Lấy thông tin bug
  SELECT SINGLE * FROM zbug_tracker INTO @ls_bug
    WHERE bug_id = @iv_bug_id AND is_del <> 'X'.
  IF sy-subrc <> 0. RETURN. ENDIF.

  " 2. Xác định recipients theo event
  CASE iv_event.
    WHEN 'CREATE'.
      SELECT * FROM zbug_users INTO TABLE @lt_recip
        WHERE role IN ('M','D') AND is_del <> 'X'.
    WHEN 'ASSIGN' OR 'REASSIGN'.
      SELECT * FROM zbug_users APPENDING TABLE @lt_recip
        WHERE user_id = @ls_bug-dev_id AND is_del <> 'X'.
    WHEN 'STATUS_CHANGE'.
      SELECT * FROM zbug_users INTO TABLE @lt_recip
        WHERE ( user_id = @ls_bug-tester_id OR user_id = @ls_bug-dev_id )
          AND is_del <> 'X'.
    WHEN 'REJECT'.
      SELECT * FROM zbug_users INTO TABLE @lt_recip
        WHERE role = 'M' AND is_del <> 'X'.
  ENDCASE.

  IF lt_recip IS INITIAL. RETURN. ENDIF.

  " 3. Tạo email body (HTML)
  lv_subject = |[BugTracker] { iv_event }: { ls_bug-bug_id } - { ls_bug-title }|.

  ls_body-line = |<html><body>|.                            APPEND ls_body TO
lt_body.
  ls_body-line = |<h2>Bug Tracking Notification</h2>|.     APPEND ls_body TO
lt_body.
  ls_body-line = |<p><b>Event:</b> { iv_event }</p>|.      APPEND ls_body TO
lt_body.
```

```
  ls_body-line = |<p><b>Bug ID:</b> { ls_bug-bug_id }</p>|. APPEND ls_body TO
lt_body.
  ls_body-line = |<p><b>Title:</b> { ls_bug-title }</p>|.  APPEND ls_body TO
lt_body.
  ls_body-line = |<p><b>Project:</b> { ls_bug-project_id }</p>|. APPEND ls_body
TO lt_body.
  ls_body-line = |<p><b>Status:</b> { ls_bug-status }</p>|. APPEND ls_body TO
lt_body.
  ls_body-line = |<p><b>Developer:</b> { ls_bug-dev_id }</p>|. APPEND ls_body TO
lt_body.
  ls_body-line = |<p><b>Tester:</b> { ls_bug-tester_id }</p>|. APPEND ls_body TO
lt_body.
  ls_body-line = |</body></html>|.                          APPEND ls_body TO
lt_body.

  " 4. Gửi email qua CL_BCS
  TRY.
    lo_send_req = cl_bcs=>create_persistent( ).
    lo_doc = cl_document_bcs=>create_document(
      i_type    = 'HTM'
      i_text    = lt_body
      i_subject = lv_subject ).
    lo_send_req->set_document( lo_doc ).

    lo_sender = cl_sapuser_bcs=>create( sy-uname ).
    lo_send_req->set_sender( lo_sender ).

    LOOP AT lt_recip ASSIGNING FIELD-SYMBOL(<user>).
      IF <user>-email IS NOT INITIAL.
        lo_recip = cl_cam_address_bcs=>create_internet_address( <user>-email ).
        lo_send_req->add_recipient( lo_recip ).
      ENDIF.
    ENDLOOP.

    lo_send_req->set_send_immediately( 'X' ).
    lv_sent = lo_send_req->send( ).
    COMMIT WORK.
  CATCH cx_bcs INTO DATA(lx_bcs).
    " Log error but don't block main flow
    DATA(lv_err) = lx_bcs->get_text( ).
  ENDTRY.

ENDFUNCTION.
```

Bước B7: Cập nhật Soft Delete Logic

Mục tiêu: Đảm bảo hệ thống không xóa thực sự dữ liệu (Hard Delete) mà chỉ đánh dấu xóa (Soft Delete) qua field `IS_DEL = 'X'`.

Thao tác thực hiện: Bạn cần vào **SE37**, mở lần lượt 2 FM sau đây để sửa code:

1. FM `Z_BUG_GET` (Lấy thông tin Bug)

- **Mục tiêu:** Thêm điều kiện lọc `is_del <> 'X'`.
- **Source code:**

```
SELECT SINGLE * FROM zbug_tracker INTO @es_bug
  WHERE bug_id = @iv_bug_id
    AND is_del <> 'X'.

IF sy-subrc = 0.
  ev_success = 'Y'.
ELSE.
  ev_success = 'N'.
  ev_message = 'Bug not found or was deleted'.
ENDIF.
```

2. FM `Z_BUG_DELETE` (Đánh dấu xóa)

- **Mục tiêu:** Thay vì `DELETE`, ta dùng `UPDATE` field `IS_DEL`.
- **Source code:**

```
DATA: ls_bug TYPE zbug_tracker.

" 1. Kiểm tra bug tồn tại
SELECT SINGLE * FROM zbug_tracker INTO @ls_bug
  WHERE bug_id = @iv_bug_id
    AND is_del <> 'X'.

IF sy-subrc <> 0.
  ev_success = 'N'.
  ev_message = 'Bug not found or already deleted'.
  RETURN.
ENDIF.

" 2. Thực hiện Soft Delete (Dùng @ và dấu phẩy cho Strict Mode)
UPDATE zbug_tracker
  SET is_del = 'X',
      aenam  = @sy-uname,
      aedat  = @sy-datum,
      aezet  = @sy-uzeit
  WHERE bug_id = @iv_bug_id.

IF sy-subrc = 0.
  COMMIT WORK.
  ev_success = 'Y'.
  CONCATENATE 'Bug' iv_bug_id 'marked as deleted' INTO ev_message SEPARATED BY
SPACE.
ELSE.
```

```
    ROLLBACK WORK.
    ev_success = 'N'.
    ev_message = 'Failed to delete bug'.
  ENDIF.
```

Nhấn **Activate** cho cả 2 FM.

> ✅ **Checkpoint:** Bất kỳ query nào `SELECT FROM zbug_tracker/zbug_users` đều có `WHERE is_del <> 'X'`.

Bước B8: Cập nhật `Z_BUG_AUTO_ASSIGN`

Mục tiêu: Update FM auto-assign để filter IS_DEL + chỉ assign dev trong cùng project.

> ⚠️ FM `Z_BUG_AUTO_ASSIGN` đã tồn tại từ MVP nhưng cần update đáng kể.

Vào **SE37** → mở FM `Z_BUG_AUTO_ASSIGN` → **Change**.

Source code cập nhật (thay thế toàn bộ):

```
FUNCTION z_bug_auto_assign.
*"----------------------------------------------------------------------
*"*"Local Interface:
*"  IMPORTING
*"     VALUE(IV_BUG_ID) TYPE  ZDE_BUG_ID
*"     VALUE(IV_MODULE) TYPE  ZDE_SAP_MODULE
*"  EXPORTING
*"     VALUE(EV_DEV_ID) TYPE  ZDE_USERNAME
*"     VALUE(EV_STATUS) TYPE  ZDE_BUG_STATUS
*"     VALUE(EV_MESSAGE) TYPE  STRING
*"----------------------------------------------------------------------

  DATA: ls_bug      TYPE zbug_tracker,
        lt_devs     TYPE TABLE OF zbug_users,
        ls_dev      TYPE zbug_users,
        lv_count    TYPE i,
        lv_min      TYPE i VALUE 99999,
        lv_best_dev TYPE zde_username,
        lv_uname    TYPE sy-uname.

  lv_uname = sy-uname.

  " 1. Lấy thông tin bug
  SELECT SINGLE * FROM zbug_tracker INTO @ls_bug
    WHERE bug_id = @iv_bug_id AND is_del <> 'X'.
  IF sy-subrc <> 0.
    ev_status = 'N'.
```

```
  ev_message = 'Bug not found'.
  RETURN.
ENDIF.

" 2. Chỉ auto-assign cho bug ở trạng thái NEW (1)
IF ls_bug-status <> '1'.
  ev_status = 'N'.
  ev_message = 'Bug is not in NEW status'.
  RETURN.
ENDIF.

" 3. Tìm các Dev trong Project (ZBUG_USER_PROJEC)
SELECT u~user_id FROM zbug_users AS u
  INNER JOIN zbug_user_projec AS up ON u~user_id = up~user_id
  INTO TABLE @lt_devs
  WHERE u~role = 'D'
    AND u~is_del <> 'X'
    AND up~project_id = @ls_bug-project_id.

IF lt_devs IS INITIAL.
  " Không có Dev -> Chuyển sang trạng thái Waiting (W)
  UPDATE zbug_tracker SET status = 'W', aenam = @lv_uname
    WHERE bug_id = @iv_bug_id.
  COMMIT WORK.
  ev_status = 'W'.
  ev_message = 'No dev available - status set to Waiting'.
  RETURN.
ENDIF.

" 4. Tìm Dev ít việc nhất (Round-robin)
LOOP AT lt_devs INTO ls_dev.
  SELECT COUNT(*) FROM zbug_tracker INTO @lv_count
    WHERE dev_id = @ls_dev-user_id
      AND is_del <> 'X'
      AND status IN ('2','3').
  IF lv_count < lv_min.
    lv_min = lv_count.
    lv_best_dev = ls_dev-user_id.
  ENDIF.
ENDLOOP.

" 5. Cập nhật Assign
UPDATE zbug_tracker SET dev_id = @lv_best_dev,
                        status = '2',
                        aenam  = @lv_uname
  WHERE bug_id = @iv_bug_id.

IF sy-subrc = 0.
  COMMIT WORK.
```

```
    ev_status = '2'. " Assigned
    ev_dev_id = lv_best_dev.
    ev_message = 'Auto-assigned successfully'.

    " 6. Gửi Email thông báo
    CALL FUNCTION 'Z_BUG_SEND_EMAIL'
      EXPORTING
        iv_bug_id = iv_bug_id
        iv_event  = 'ASSIGN'.
  ENDIF.

ENDFUNCTION.
```

Nhấn **Activate**.

> ✅ **Checkpoint: SE37** → `Z_BUG_AUTO_ASSIGN`:

- Bug trong project có 1 dev → auto-assign, STATUS=2
- Bug trong project có 0 dev → STATUS='W' (Waiting)
- Dev đã bị soft-delete → không được assign

Bước B9: Tạo mới `Z_BUG_REASSIGN`

Mục tiêu: FM mới cho phép Manager chuyển bug sang dev khác.

> ⚠️ FM này **KHÔNG tồn tại** trong MVP — cần tạo hoàn toàn mới trong Function Group `ZBUG_FG`.

Vào **SE37** → nhập `Z_BUG_REASSIGN` → **Create** → Function Group `ZBUG_FG`.

Source code:

```
FUNCTION z_bug_reassign.
*"----------------------------------------------------------------------
*"*"Local Interface:
*"  IMPORTING
*"     VALUE(IV_BUG_ID) TYPE  ZDE_BUG_ID
*"     VALUE(IV_NEW_DEV_ID) TYPE  ZDE_USERNAME
*"     VALUE(IV_REASON) TYPE  ZDE_REASONS
*"     VALUE(IV_REQUESTED_BY) TYPE  ZDE_USERNAME
*"  EXPORTING
*"     VALUE(EV_STATUS) TYPE  ZDE_BUG_STATUS
*"     VALUE(EV_MESSAGE) TYPE  STRING
*"----------------------------------------------------------------------

  DATA: ls_bug      TYPE zbug_tracker,
```

```
      lv_role     TYPE char1,
      lv_count    TYPE i,
      lv_uname    TYPE sy-uname.

lv_uname = sy-uname.

" 1. Kiểm tra Bug tồn tại
SELECT SINGLE * FROM zbug_tracker INTO @ls_bug
  WHERE bug_id = @iv_bug_id AND is_del <> 'X'.
IF sy-subrc <> 0.
  ev_status = 'N'.
  ev_message = 'Bug not found'.
  RETURN.
ENDIF.

" 2. Kiểm tra quyền Manager
SELECT SINGLE role FROM zbug_users INTO @lv_role
  WHERE user_id = @lv_uname AND is_del <> 'X'.
IF lv_role <> 'M'.
  ev_status = 'N'.
  ev_message = 'Only Manager can reassign'.
  RETURN.
ENDIF.

" 3. Kiểm tra Dev mới có thuộc Project không (ZBUG_USER_PROJEC)
SELECT COUNT(*) FROM zbug_user_projec INTO @lv_count
  WHERE user_id = @iv_new_dev_id
    AND project_id = @ls_bug-project_id.
IF lv_count = 0.
  ev_status = 'N'.
  ev_message = 'New developer must belong to same project'.
  RETURN.
ENDIF.

" 4. Cập nhật Dev mới
UPDATE zbug_tracker SET dev_id = @iv_new_dev_id,
                        status = '2',
                        aenam  = @lv_uname
  WHERE bug_id = @iv_bug_id.

IF sy-subrc = 0.
  ev_status = '2'.
  COMMIT WORK.

  " Gửi Email thông báo REASSIGN
  CALL FUNCTION 'Z_BUG_SEND_EMAIL'
    EXPORTING
      iv_bug_id = iv_bug_id
      iv_event  = 'REASSIGN'.
```

```
  ELSE.
    ev_status = 'N'.
    ROLLBACK WORK.
  ENDIF.

ENDFUNCTION.
```

Nhấn **Activate**.

✅ **Checkpoint: SE37** → `Z_BUG_REASSIGN`:

- Manager reassign từ Dev1 sang Dev2 (cùng project) → Success, DEV_ID=Dev2
- Non-manager gọi → "Only Manager…"
- Dev2 không thuộc project → "New developer must belong to the same project"
- Bug ở status R → status reset về 2 (Assigned)

TỔNG KẾT PHASE B

Sau khi hoàn thành Phase B, bạn phải có:

- ☒ `Z_BUG_CHECK_PERMISSION` — hỗ trợ 12 actions (Bug + Project) + project membership
- ☒ `Z_BUG_CREATE` — validate Project + Severity + User membership + Config bug branch
- ☒ `Z_BUG_UPDATE_STATUS` — 9 status states + transition validation + evidence check + special fields (CLOSED_AT, VERIFY_TESTER_ID, APPROVED_BY)
- ☒ `Z_BUG_GOS_UPLOAD` / `Z_BUG_GOS_LIST` — upload/list file qua BDS
- ☒ `ZBUG_EMAIL_FORM` — SmartForm template HTML
- ☒ `Z_BUG_SEND_EMAIL` — CL_BCS + HTML + event-based recipients (CREATE, ASSIGN, REASSIGN, STATUS_CHANGE, REJECT)
- ☒ Soft delete logic áp dụng trong tất cả các FM
- ☒ `Z_BUG_AUTO_ASSIGN` — **updated**: IS_DEL filter + project membership + Waiting status fallback
- ☒ `Z_BUG_REASSIGN` — **MỚI**: Manager reassign dev + status reset + validation

Tổng FM count: 9 FMs (7 updated + 2 mới: GOS_LIST, REASSIGN)

👉 **Chuyển sang Phase C: Module Pool UI** # HƯỚNG DẪN TRIỂN KHAI CHI TIẾT — PHASE C: MODULE POOL UI (v2.0)

Dự án: SAP Bug Tracking Management System **Cập nhật:** 09/04/2026 | **Phiên bản:** 2.0 (Project-First Flow) **SAP:** S40 | Client 324 | **ABAP 770 Package:** `ZBUGTRACK` | **Program:** `Z_BUG_WORKSPACE_MP` (Type M) **Yêu cầu:** Phase A (database) hoàn thành, Phase B (business logic) đã có FMs

BREAKING CHANGES so với v1.0: - Screen 0400 (Project List) là **initial screen** — thay vì Screen 0100 (Hub) - Screen 0100 (Hub) **deprecated** — code giữ nhưng không navigate tới - Bug **bắt buộc thuộc 1 Project** — không cho tạo bug lỏng - Project hotspot trên Project ALV → **mở Bug List** (thay vì Project Detail) - Nút **My Bugs** trên Project List → xem bugs cross-project theo role - Description mini editor trên Bug Info tab (subscreen 0310) - Type fixes: STATUS = CHAR 20, SAP_MODULE = CHAR 20, REASON = STRING

C1: Tạo Program + Includes

Skip nếu đã tạo — program `Z_BUG_WORKSPACE_MP` + 6 includes đã tồn tại.

SE80 Steps:

1. **SE80** → chọn "Program" → nhập `Z_BUG_WORKSPACE_MP` → Enter
2. "With TOP INCL." = Yes → Type = **Module Pool (M)** → Status = Test (T) → Package = `ZBUGTRACK`
3. SAP tự tạo `Z_BUG_WS_TOP`
4. Right-click program → Create → Include → lần lượt tạo:
   - `Z_BUG_WS_F00`
   - `Z_BUG_WS_PBO`
   - `Z_BUG_WS_PAI`
   - `Z_BUG_WS_F01`
   - `Z_BUG_WS_F02`

Main Program Code:

```
PROGRAM z_bug_workspace_mp.
INCLUDE z_bug_ws_top.    " 1. Global data
INCLUDE z_bug_ws_f00.    " 2. Event class (PHẢI trước PBO/PAI)
INCLUDE z_bug_ws_pbo.    " 3. PBO
INCLUDE z_bug_ws_pai.    " 4. PAI
INCLUDE z_bug_ws_f01.    " 5. Business logic
INCLUDE z_bug_ws_f02.    " 6. Helpers
```

Thứ tự bắt buộc: F00 TRƯỚC PBO/PAI vì class `lcl_event_handler` cần define trước khi reference.

Save + Activate.

Checkpoint: SE80 → thấy 6 includes trong navigation tree.

C2: Copy Code vào 6 Includes

Mở từng include trong SE80, **xóa hết nội dung cũ**, paste code mới từ các file guide:

Include	Copy từ file	Nội dung
`Z_BUG_WS_TOP`	`CODE_TOP.md`	Global vars, types, constants, ALV objects
`Z_BUG_WS_F00`	`CODE_F00.md`	Event handler class + 3 field catalogs
`Z_BUG_WS_PBO`	`CODE_PBO.md`	11 PBO modules
`Z_BUG_WS_PAI`	`CODE_PAI.md`	5 PAI modules + table control sync
`Z_BUG_WS_F01`	`CODE_F01.md`	15 FORM routines (SQL, save, delete, status…)
`Z_BUG_WS_F02`	`CODE_F02.md`	8 FORM routines (F4 helps, long text load/save)

Quy trình copy:

1. Mở `CODE_TOP.md` → copy toàn bộ nội dung ABAP (bỏ dòng markdown/comment đầu nếu có)
2. SE80 → double-click `Z_BUG_WS_TOP` → **Change** → Ctrl+A → Delete → Paste
3. **Save (Ctrl+S)**
4. Lặp lại cho 5 includes còn lại
5. Sau khi paste xong tất cả 6 → **Activate All** (Ctrl+Shift+F3):
   - Chọn tất cả objects → Activate
   - Nếu có warning (unused variables) → OK, bỏ qua
   - Nếu có **error** → xem Section "Troubleshooting" cuối file

Checkpoint: Program activate thành công, không có syntax error.

C3: SE93 — Đổi T-code Initial Screen

Quan trọng: T-code `ZBUG_WS` phải trỏ tới Screen **0400** (Project List) thay vì 0100.

1. **SE93** → nhập `ZBUG_WS` → **Change**
2. Field "Initial Screen": đổi từ `0100` → **`0400`**
3. Program name vẫn là `Z_BUG_WORKSPACE_MP`
4. **Save** → Activate

Nếu T-code chưa tồn tại: SE93 → `ZBUG_WS` → Create → Transaction with parameters (Dialog transaction) → Program = `Z_BUG_WORKSPACE_MP`, Screen = `0400`.

Checkpoint: Gõ `ZBUG_WS` → mở thẳng Screen 0400 (Project List).

C4: SE41 — Tạo 5 GUI Statuses

Cách vào SE41:

1. Gõ **SE41** → Enter
2. Program: `Z_BUG_WORKSPACE_MP`
3. Status: nhập tên (vd `STATUS_0100`) → **Create (F5)**
4. Short Description → Enter → màn hình vẽ nút hiện ra

Cách thêm nút:

- Click ô trống trong **Application Toolbar**
- Điền Function Code (FCode) + Text → Enter
- Trong **Function Keys** tab: gán phím tắt cho BACK/EXIT/CANC

Standard Toolbar (tích cho MỌI status):
Tất cả 5 statuses đều cần 3 nút standard:

Function Code	Gán vào phím	Mô tả
`BACK`	F3 (= Standard Back)	Back
`EXIT`	Shift+F3	Exit
`CANC`	F12	Cancel

STATUS_0100 — Hub (DEPRECATED)
Short Description: `Bug Tracking Hub (Deprecated)`

#	FCode	Text	Icon
1	`BUG_LIST`	Bug List	—
2	`PROJ_LIST`	Project List	—

Status này giữ để code không dump nếu ai đó vô tình call Screen 0100.

STATUS_0200 — Bug List
Short Description: `Bug List`

#	FCode	Text	Icon	Notes
1	`CREATE`	Create Bug	`ICON_CREATE`	Hidden: Dev role, My Bugs mode
2	`CHANGE`	Change	`ICON_CHANGE`	
3	`DISPLAY`	Display	`ICON_DISPLAY`	
4	`DELETE`	Delete	`ICON_DELETE`	Hidden: Dev, Tester, My Bugs mode
5	—	*(separator)*		
6	`REFRESH`	Refresh	`ICON_REFRESH`	

STATUS_0300 — Bug Detail
Short Description: `Bug Detail`

#	FCode	Text	Icon	Notes
1	`SAVE`	Save	`ICON_SYSTEM_SAVE`	Hidden: Display mode
2	`STATUS_CHG`	Change Status	`ICON_CHANGE`	Hidden: Create mode
3	—	*(separator)*		
4	`UP_FILE`	Upload Evidence	`ICON_IMPORT`	Hidden: Create mode
5	`UP_REP`	Upload Report	`ICON_IMPORT`	Hidden: Dev role
6	`UP_FIX`	Upload Fix	`ICON_IMPORT`	Hidden: Tester role

CRITICAL: Đảm bảo fcode `SAVE` có trong status. Thiếu `SAVE` = nút Save sẽ không hiện kể cả ở Change mode.

STATUS_0400 — Project List (INITIAL SCREEN)
Short Description: `Project List`

#	FCode	Text	Icon	Notes
1	`MY_BUGS`	My Bugs	`ICON_BIW_REPORT`	**NEW** — All roles
2	—	*(separator)*		
3	`CREA_PRJ`	Create Project	`ICON_CREATE`	Manager only
4	`CHNG_PRJ`	Change	`ICON_CHANGE`	Manager only
5	`DISP_PRJ`	Display	`ICON_DISPLAY`	All
6	`DEL_PRJ`	Delete	`ICON_DELETE`	Manager only
7	—	*(separator)*		
8	`UPLOAD`	Upload Excel	`ICON_IMPORT`	Manager only
9	`DN_TMPL`	Download Template	`ICON_EXPORT`	Manager only
10	`REFRESH`	Refresh	`ICON_REFRESH`	All

LƯU Ý: Nút `MY_BUGS` là mới — không có trong version cũ.

STATUS_0500 — Project Detail
Short Description: `Project Detail`

#	FCode	Text	Icon	Notes
1	`SAVE`	Save	`ICON_SYSTEM_SAVE`	Manager + Change/Create only
2	—	*(separator)*		
3	`ADD_USER`	Add User	`ICON_INSERT_ROW`	Manager + Change/Create only
4	`REMO_USR`	Remove User	`ICON_DELETE_ROW`	Manager + Change/Create only

Fcode `REMO_USR` (không phải `REMOVE_USER`) — giới hạn 8 ký tự cho fcode.

Save + Activate từng status.

Checkpoint: SE41 → tất cả 5 statuses hiện xanh (activated).

C5: SE41 — Tạo 5 Title Bars

Trong SE41, chuyển sang **Object Type: Title** (dropdown ở đầu màn hình):

1. Program: `Z_BUG_WORKSPACE_MP`
2. Object Type: **Title**
3. Nhập tên → Create

Title Name	Text	Used by
`TITLE_MAIN`	`&1`	Screen 0100 — nhận 1 param
`TITLE_BUGLIST`	`&1`	Screen 0200 — nhận title text
`TITLE_BUGDETAIL`	`&1`	Screen 0300 — nhận mode + bug_id
`TITLE_PROJLIST`	`&1`	Screen 0400 — nhận "Project List"
`TITLE_PRJDET`	`&1`	Screen 0500 — nhận project name

> **Cách SET TITLEBAR hoạt động:** Khi code viết `SET TITLEBAR 'TITLE_BUGLIST' WITH 'Bugs – ProjectX'`, SAP thay `&1` bằng text đó. Vì vậy chỉ cần define `&1` trong Title.

Save + Activate.

C6: Screen 0400 — Project List (INITIAL)

> **Đây là screen đầu tiên user thấy** khi mở T-code `ZBUG_WS`.

C6.1: Tạo Screen trong SE51

1. **SE80** → `Z_BUG_WORKSPACE_MP` → Right-click → Create → **Screen** → Number: **0400**
2. Short Description: `Project List`
3. Screen Type: **Normal**
4. Next Screen: `0400` (loop back to itself — navigation handled by PAI)

C6.2: Flow Logic (tab "Flow Logic")

```
PROCESS BEFORE OUTPUT.
  MODULE init_user_role.
  MODULE status_0400.
  MODULE init_project_list.

PROCESS AFTER INPUT.
  MODULE user_command_0400.
```

LƯU Ý: `init_user_role` phải chạy ở đây vì 0400 là initial screen. Module này chỉ load role 1 lần (CHECK gv_role IS INITIAL).

C6.3: Layout (tab "Layout")

Click **Layout** button → Screen Painter mở ra.

1. **Vẽ Custom Control:**
   - Menu: Edit → Create Element → Custom Control
   - Hoặc: click icon Custom Control trên toolbar → vẽ hình chữ nhật phủ gần toàn bộ screen
   - Name: **`CC_PROJECT_LIST`**
   - Size: chiếm ~90% diện tích screen (để lại chút margin trên/dưới)
2. **Không cần vẽ thêm gì khác** — tất cả buttons nằm trên GUI Status toolbar
3. Save + Activate Screen

Checkpoint: SE80 → double-click Screen 0400 → thấy `CC_PROJECT_LIST` trong Element List.

C7: Screen 0200 — Bug List (Dual Mode)

C7.1: Tạo Screen

1. SE80 → Right-click program → Create → Screen → **0200**
2. Short Description: `Bug List`
3. Screen Type: Normal
4. Next Screen: `0200`

C7.2: Flow Logic

```
PROCESS BEFORE OUTPUT.
  MODULE status_0200.
  MODULE init_bug_list.

PROCESS AFTER INPUT.
  MODULE user_command_0200.
```

C7.3: Layout

1. Vẽ Custom Control: **`CC_BUG_LIST`** — phủ gần toàn bộ screen
2. Save + Activate

Checkpoint: Screen 0200 activated.

C8: Screen 0300 — Bug Detail (Tab Strip Host)

C8.1: Tạo Screen

1. SE80 → Create Screen → **0300**
2. Short Description: `Bug Detail`
3. Screen Type: Normal
4. Next Screen: `0300`

C8.2: Flow Logic

```
PROCESS BEFORE OUTPUT.
  MODULE status_0300.
  MODULE load_bug_detail.
  MODULE modify_screen_0300.
  CALL SUBSCREEN ss_tab INCLUDING sy-repid gv_active_subscreen.

PROCESS AFTER INPUT.
  CALL SUBSCREEN ss_tab.
  MODULE user_command_0300.
```

> **CALL SUBSCREEN** yêu cầu: - Có Subscreen Area tên `SS_TAB` trên layout - `gv_active_subscreen` chứa screen number (0310-0360) - PBO: `INCLUDING sy-repid gv_active_subscreen` - PAI: chỉ `CALL SUBSCREEN ss_tab.` (không có INCLUDING)

C8.3: Layout — Tab Strip + Subscreen Area

Bước 1: Vẽ Tab Strip

1. Mở Layout Editor
2. Menu: Edit → Create Element → Tab Strip (hoặc click icon TabStrip trên toolbar)
3. Vẽ hình chữ nhật phủ ~85% screen (để chừa toolbar phía trên)
4. Name: **`TS_DETAIL`**
5. SAP hỏi số tabs → nhập **6**
6. SAP tạo 6 tab buttons. Đặt tên + FCode cho từng tab:

Tab #	Name (button)	Text	FCode
1	`TAB_INFO`	Bug Info	`TAB_INFO`
2	`TAB_DESC`	Description	`TAB_DESC`
3	`TAB_DEVNOTE`	Dev Note	`TAB_DEVNOTE`
4	`TAB_TSTR_NOTE`	Tester Note	`TAB_TSTR_NOTE`
5	`TAB_EVIDENCE`	Evidence	`TAB_EVIDENCE`
6	`TAB_HISTORY`	History	`TAB_HISTORY`

Bước 2: Vẽ Subscreen Area bên trong Tab Strip

1. Click vào vùng trống bên trong tab strip body
2. Menu: Edit → Create Element → Subscreen Area
3. Vẽ hình chữ nhật lấp đầy phần body của tab strip
4. Name: **`SS_TAB`**

Bước 3: Kết nối Tab Strip với subscreen (Reference Field)

1. Double-click vào tab strip `TS_DETAIL` → Attributes
2. Đảm bảo "Control" Reference: `TS_DETAIL` (trong TOP đã có `CONTROLS: ts_detail TYPE TABSTRIP`)
3. Mỗi tab button có FCode tương ứng — PAI sẽ handle switch

Save + Activate Screen 0300.

Checkpoint: Screen 0300 có Tab Strip với 6 tabs + Subscreen Area `SS_TAB`.

C9: Subscreens 0310-0360

> **Tất cả subscreens** phải có Screen Type = **Subscreen**.

C9.1: Screen 0310 — Bug Info (Fields + Description Mini Editor)

Tạo Screen: 1. SE80 → Create Screen → **0310** 2. Short Description: `Bug Info` 3. Screen Type: **Subscreen**

Flow Logic:

```
PROCESS BEFORE OUTPUT.
  MODULE init_desc_mini.

PROCESS AFTER INPUT.
```

> PBO module `init_desc_mini` tạo/load description mini editor.

Layout — Step by step:

Bước 1: Thêm fields từ Dictionary

1. Mở Layout Editor → click **Dict/Program Fields** (icon book trên toolbar)
2. Table/Field Name: `GS_BUG_DETAIL` → Get from Program
3. SAP liệt kê tất cả fields của `gs_bug_detail` (work area TYPE zbug_tracker)
4. Tick chọn các fields cần hiện:
   - `BUG_ID`, `TITLE`, `PROJECT_ID`, `STATUS`, `PRIORITY`
   - `SEVERITY`, `BUG_TYPE`, `SAP_MODULE`
   - `TESTER_ID`, `DEV_ID`, `VERIFY_TESTER_ID`
   - `CREATED_AT`

5. Click **Enter** → SAP tạo labels + input fields → kéo thả sắp xếp

Bước 2: Thêm display text fields

1. Dict/Program Fields → Variable: `GV_STATUS_DISP` → Get from Program
2. Kéo field ra cạnh STATUS field → Set **Input = OFF** (display-only)
3. Lặp lại cho: `GV_PRIORITY_DISP`, `GV_SEVERITY_DISP`, `GV_BUG_TYPE_DISP`

Bước 3: Set Screen Groups

Double-click từng field → tab Attributes → set Group1:

Field	Group1	Purpose
`GS_BUG_DETAIL-BUG_ID`	**BID**	Display-only after creation
`GS_BUG_DETAIL-PROJECT_ID`	**PRJ**	Locked when creating from project
`GS_BUG_DETAIL-TITLE`	**EDT**	Editable (disabled in Display mode)
`GS_BUG_DETAIL-STATUS`	**EDT**	Editable
`GS_BUG_DETAIL-PRIORITY`	**EDT**	Editable
`GS_BUG_DETAIL-SEVERITY`	**EDT**	Editable
`GS_BUG_DETAIL-BUG_TYPE`	**EDT**	Editable
`GS_BUG_DETAIL-SAP_MODULE`	**EDT**	Editable
`GS_BUG_DETAIL-TESTER_ID`	**TST**	Only Tester/Manager can edit
`GS_BUG_DETAIL-VERIFY_TESTER_ID`	**TST**	Only Tester/Manager can edit
`GS_BUG_DETAIL-DEV_ID`	**DEV**	Only Dev/Manager can edit
`GS_BUG_DETAIL-CREATED_AT`	—	Always display-only (Input = OFF)
`GV_STATUS_DISP`	—	Always display-only
`GV_PRIORITY_DISP`	—	Always display-only
`GV_SEVERITY_DISP`	—	Always display-only
`GV_BUG_TYPE_DISP`	—	Always display-only

Bước 4: Thêm Group Boxes (optional UX)

1. Menu: Edit → Create Element → Box
2. Vẽ box quanh nhóm fields → set Text:
   - **"Bug Information"** — quanh BUG_ID, TITLE, PROJECT_ID, STATUS, PRIORITY
   - **"Classification"** — quanh SEVERITY, BUG_TYPE, SAP_MODULE
   - **"Assignment"** — quanh TESTER_ID, DEV_ID, VERIFY_TESTER_ID
   - **"Description"** — quanh CC_DESC_MINI

Bước 5: Thêm Description Mini Editor (Custom Control)

1. Edit → Create Element → Custom Control
2. Vẽ hình chữ nhật nhỏ (~60 chars wide x 4 lines high) ở phần dưới screen
3. Name: **`CC_DESC_MINI`**

Bước 6: Labels

Rename labels cho rõ ràng: - `BUG_ID` → "Bug ID" - `TITLE` → "Title *" (required indicator)* - *`PROJECT_ID`* → *"Project* " - `STATUS` → "Status" - etc.

Save + Activate Screen 0310.

Layout Preview (approximate):

```
┌─ Bug Information ──────────────────────────────────────────┐
│ Bug ID:        [__________]                                │
│ Title *:       [_________________________________________]  │
│ Project *:     [___________________]                       │
│ Status:        [____] → [New         ] (display)           │
│ Priority:      [_]   → [Medium      ] (display)           │
└────────────────────────────────────────────────────────────┘

┌─ Classification ───────────────────────────────────────────┐
│ Severity:      [_]   → [Normal      ] (display)           │
│ Bug Type:      [_]   → [Functional  ] (display)           │
│ SAP Module:    [___________________]                       │
└────────────────────────────────────────────────────────────┘

┌─ Assignment ───────────────────────────────────────────────┐
│ Tester:        [_____________]                             │
│ Developer:     [_____________]                             │
│ Verify Tester: [_____________]                             │
│ Created Date:  [__________] (display-only)                 │
└────────────────────────────────────────────────────────────┘

┌─ Description ──────────────────────────────────────────────┐
│ ┌──────────────────────────────────────────────────────┐   │
│ │ (CC_DESC_MINI — cl_gui_textedit, 3-4 lines)         │  │
│ │                                                      │   │
│ └──────────────────────────────────────────────────────┘   │
└────────────────────────────────────────────────────────────┘
```

C9.2: Screen 0320 — Description (Long Text Z001)

1. Create Screen **0320**, Type = **Subscreen**
2. Flow Logic: empty (PBO/PAI trống)

```
PROCESS BEFORE OUTPUT.
PROCESS AFTER INPUT.
```

3. Layout: vẽ Custom Control **`CC_DESC`** chiếm toàn bộ subscreen
4. Save + Activate

> Long Text editor (cl_gui_textedit) được tạo trong PBO khi tab được click. Xem `load_long_text` trong CODE_PAI.md → call khi user chọn TAB_DESC.

C9.3: Screen 0330 — Dev Note (Long Text Z002)

1. Create Screen **0330**, Type = Subscreen
2. Flow Logic: empty
3. Layout: Custom Control **`CC_DEV_NOTE`**
4. Save + Activate

C9.4: Screen 0340 — Tester Note (Long Text Z003)

1. Create Screen **0340**, Type = Subscreen
2. Flow Logic: empty
3. Layout: Custom Control **`CC_TSTR_NOTE`**
4. Save + Activate

C9.5: Screen 0350 — Evidence (GOS)

1. Create Screen **0350**, Type = Subscreen
2. Flow Logic: empty
3. Layout: Custom Control **`CC_EVIDENCE`**
4. Save + Activate

> Phase D: GOS attachment logic sẽ dùng container `CC_EVIDENCE`.

C9.6: Screen 0360 — History (ALV readonly)

1. Create Screen **0360**, Type = Subscreen
2. Flow Logic: empty
3. Layout: Custom Control **`CC_HISTORY`** chiếm toàn bộ subscreen
4. Save + Activate

History ALV được tạo trong `load_history_data` (CODE_F01.md) khi user click tab History.

C10: Screen 0500 — Project Detail + Table Control

C10.1: Tạo Screen

1. SE80 → Create Screen → **0500**
2. Short Description: `Project Detail`
3. Screen Type: Normal
4. Next Screen: `0500`

C10.2: Flow Logic

```
PROCESS BEFORE OUTPUT.
  MODULE status_0500.
  MODULE init_project_detail.
  MODULE modify_screen_0500.
  LOOP AT gt_user_project INTO gs_user_project WITH CONTROL tc_users.
  ENDLOOP.

PROCESS AFTER INPUT.
  LOOP AT gt_user_project.
    MODULE tc_users_modify ON CHAIN-REQUEST.
  ENDLOOP.
  MODULE user_command_0500.
```

C10.3: Layout

Bước 1: Thêm Project fields

1. Dict/Program Fields → `GS_PROJECT` → Get from Program
2. Chọn: `PROJECT_ID`, `PROJECT_NAME`, `DESCRIPTION`, `PROJECT_STATUS`, `START_DATE`, `END_DATE`, `PROJECT_MANAGER`, `NOTE`
3. Kéo thả sắp xếp thành 2 cột

Bước 2: Thêm display text cho Project Status

1. Dict/Program Fields → `GV_PRJ_STATUS_DISP` → Get from Program
2. Đặt cạnh `GS_PROJECT-PROJECT_STATUS` → Input = OFF (display-only)

Bước 3: Set Screen Groups

Field	Group1
`GS_PROJECT-PROJECT_ID`	**EDT**
`GS_PROJECT-PROJECT_NAME`	**EDT**
`GS_PROJECT-DESCRIPTION`	**EDT**
`GS_PROJECT-PROJECT_STATUS`	**EDT**
`GS_PROJECT-START_DATE`	**EDT**
`GS_PROJECT-END_DATE`	**EDT**
`GS_PROJECT-PROJECT_MANAGER`	**EDT**
`GS_PROJECT-NOTE`	**EDT**
`GV_PRJ_STATUS_DISP`	— (display-only)

Bước 4: Vẽ Table Control

1. Edit → Create Element → Table Control
2. Vẽ hình chữ nhật ở phần dưới screen (~50% diện tích)
3. Name: **`TC_USERS`**
4. SAP hỏi columns → thêm columns:

Column	Field Name	Header Text	Width
1	`GS_USER_PROJECT-USER_ID`	User ID	12
2	`GS_USER_PROJECT-ROLE`	Role	5
3	`GS_USER_PROJECT-ERNAM`	Created By	12
4	`GS_USER_PROJECT-ERDAT`	Created On	10

Cách thêm columns vào Table Control:

1. Double-click Table Control `TC_USERS` → Attributes mở ra
2. Hoặc: click vào bên trong table control → Dict/Program Fields → `GS_USER_PROJECT` → chọn `USER_ID`, `ROLE`, `ERNAM`, `ERDAT` → kéo vào table control

Bước 5: Labels

- Add label “Team Members” phía trên Table Control
- PROJECT_ID → “Project ID *”
- PROJECT_NAME → “Project Name *”

Save + Activate Screen 0500.

Layout Preview:

```
┌─ Project Information ──────────────────────────────────────┐
│ Project ID *:  [__________________]                         │
│ Project Name *:[_________________________________________] │
│ Description:   [_________________________________________] │
│ Status:        [_] → [Opening     ] (display)               │
│ Start Date:    [__________]   End Date: [__________]        │
│ Manager:       [_____________]                              │
│ Note:          [_________________________________________] │
└────────────────────────────────────────────────────────────┘

Team Members:
┌──────────────┬──────┬──────────────┬──────────┐
│ User ID      │ Role │ Created By   │ Created  │
├──────────────┼──────┼──────────────┼──────────┤
│ DEV-089      │ M    │ DEV-089      │ 09.04.26 │
│ DEV-061      │ D    │ DEV-089      │ 09.04.26 │
│ DEV-118      │ T    │ DEV-089      │ 09.04.26 │
└──────────────┴──────┴──────────────┴──────────┘
```

C11: Screen 0100 — Hub (DEPRECATED)

Screen 0100 **không được sử dụng** trong flow mới. Giữ lại để tránh dump.

Tạo Screen (nếu chưa tạo):

1. SE80 → Create Screen → **0100**
2. Short Description: `Hub (DEPRECATED)`
3. Screen Type: Normal

Flow Logic:

```
PROCESS BEFORE OUTPUT.
  MODULE status_0100.
  MODULE init_user_role.

PROCESS AFTER INPUT.
  MODULE user_command_0100.
```

Layout:

- Trống hoặc 1 text "This screen is deprecated. Use Project List instead."
- Không cần buttons — PAI chỉ handle BACK/EXIT/CANC → LEAVE PROGRAM

Save + Activate.

C12: Deprecate Old Programs

Sau khi Module Pool hoạt động, đánh dấu SE38 programs cũ:

1. SE38 → mỗi program cũ → Attributes → Title thêm `[DEPRECATED]`
2. Thêm comment đầu program:

```
* ============================================
* [DEPRECATED] — Replaced by Z_BUG_WORKSPACE_MP
* Use T-code ZBUG_WS instead.
* ============================================
```

Old Program	Old T-Code	Replaced by
`Z_BUG_CREATE_SCREEN`	`ZBUG_CREATE`	Screen 0300 (Create mode)
`Z_BUG_UPDATE_SCREEN`	`ZBUG_UPDATE`	Screen 0300 (Change mode)
`Z_BUG_REPORT_ALV`	`ZBUG_REPORT`	Screen 0200 (Bug List ALV)
`Z_BUG_MANAGER_DASHBOARD`	`ZBUG_MANAGER`	Screen 0400 (Project List)
`Z_BUG_PRINT`	`ZBUG_PRINT`	Screen 0200 (Print button)

KHÔNG XÓA programs cũ — chỉ deprecate. Xóa sau Phase E testing.

C13: Testing Checklist

Flow Tests:

- ☐ `ZBUG_WS` → mở Screen 0400 (Project List)
- ☐ Click project hotspot → mở Screen 0200 (Bug List, all bugs of project)
- ☐ "My Bugs" button → mở Screen 0200 (filtered by role)
- ☐ Back từ Bug List → Screen 0400
- ☐ Back từ Project List → LEAVE PROGRAM
- ☐ Create Bug (từ project) → Screen 0300 (PROJECT_ID pre-filled + locked)
- ☐ Create Bug button hidden trong My Bugs mode
- ☐ Back từ Bug Detail → Screen 0200

CRUD Tests:

- ☐ Create Bug → auto-gen BUG_ID → save → status changes to "Change" mode
- ☐ BUG_ID display-only after creation
- ☐ Change Bug → modify fields → Save successful
- ☐ Display Bug → all fields readonly
- ☐ Delete Bug → confirm popup → soft delete → ALV refreshes
- ☐ Create Project (Manager only) → save → success
- ☐ Add User to Project → popup → save → table control updates

- ☐ Remove User from Project → confirm → success

Display Tests:

- ☐ Status shows text (New/Assigned/…) not raw code (1/2/…)
- ☐ Priority shows text (High/Medium/Low) not raw code (H/M/L)
- ☐ Severity shows text in ALV (Dump/Critical, Very High, …)
- ☐ Bug Type shows text in ALV (Functional, Performance, …)
- ☐ Project Status shows text (Opening/In Process/…)
- ☐ Title bar shows mode: "Create Bug" / "Change Bug: BUG0000001"
- ☐ ALV rows color-coded by status

Role Tests:

- ☐ Tester: cannot create bugs, cannot delete bugs
- ☐ Developer: cannot create bugs, cannot delete bugs, cannot edit Tester fields
- ☐ Manager: full access
- ☐ Tester: cannot see Dev-only fields (DEV group disabled)
- ☐ Non-manager: cannot create/change/delete projects

Tab Tests:

- ☐ Tab Info → fields + description mini editor visible
- ☐ Tab Description → long text editor (Z001)
- ☐ Tab Dev Note → long text editor (Z002)
- ☐ Tab Tester Note → long text editor (Z003)
- ☐ Tab Evidence → placeholder (Phase D)
- ☐ Tab History → ALV with change log

TROUBLESHOOTING

"Screen field not found" error

Nguyên nhân: Layout reference field tên khác với code (vd `GS_BUG_DETAIL-BUG_ID` vs `GS_BUG-BUG_ID`). **Fix:** Trong SE51 Layout, double-click field → kiểm tra name khớp chính xác với `GS_BUG_DETAIL-xxx` (global work area).

"Module xxx not found" error

Nguyên nhân: Flow Logic reference module chưa tồn tại trong PBO/PAI include. **Fix:** Kiểm tra tên module trong Flow Logic khớp với tên trong `Z_BUG_WS_PBO/Z_BUG_WS_PAI`. Module Pool không cho phép typo.

“Class lcl_event_handler unknown” error

Nguyên nhân: Include `Z_BUG_WS_F00` nằm SAU `PBO/PAI` trong main program. **Fix:** Đảm bảo thứ tự include: TOP → **F00** → PBO → PAI → F01 → F02.

ALV không hiện data

Nguyên nhân: Custom Container name trên screen khác với code. **Fix:** Layout có `CC_BUG_LIST` → code có `container_name = 'CC_BUG_LIST'`. Phải khớp chính xác (case-insensitive).

Tab strip không switch

Nguyên nhân: Subscreen Area name không khớp hoặc FCode tab buttons chưa set. **Fix:** Double-click mỗi tab button → Attributes → Function Code phải = `TAB_INFO`/`TAB_DESC`/etc.

Save button không hiện

Nguyên nhân: Fcode `SAVE` thiếu trong GUI Status `STATUS_0300`. **Fix:** SE41 → `STATUS_0300` → thêm `SAVE` vào Application Toolbar.

TỔNG KẾT PHASE C (v2.0)

Sau khi hoàn thành:

- ☒ Program `Z_BUG_WORKSPACE_MP` + 6 includes (code mới v2.0)
- ☒ T-code `ZBUG_WS` → Screen **0400** (thay vì 0100)
- ☒ Screen 0400 — Project List (initial, ALV, My Bugs button)
- ☒ Screen 0200 — Bug List (dual mode: Project / My Bugs)
- ☒ Screen 0300 — Bug Detail (Tab Strip, 6 subscreens)
- ☒ Subscreen 0310 — Bug Info + Description Mini Editor + Group Boxes
- ☒ Subscreens 0320-0360 — Long Text / Evidence / History
- ☒ Screen 0500 — Project Detail + Table Control (with ROLE column)
- ☒ Screen 0100 — Hub (deprecated, kept for safety)
- ☒ 5 GUI Statuses with correct fcodes (MY_BUGS, REMO_USR, etc.)
- ☒ 5 Title Bars with dynamic `&1` parameter
- ☒ Role-based button exclusion (Manager/Tester/Dev)
- ☒ Screen groups: EDT, BID, PRJ, TST, DEV
- ☒ Display text fields: status/priority/severity/bug_type/project_status
- ☒ ALV: severity_text, bug_type_text columns (raw hidden)
- ☒ Old programs deprecated

Next: Phase D — Excel Upload, GOS Attachments, Message Class Migration # HƯỚNG DẪN TRIỂN KHAI CHI TIẾT — PHASE D: EXCEL & ADVANCED FEATURES

Dự án: SAP Bug Tracking Management System **Ngày:** 09/04/2026 | **Phiên bản:** 6.0 (Project-First Flow) **Thời gian ước tính:** 2 ngày **Yêu cầu:** Hoàn thành Phase A + B + C trước **ABAP Version:** 7.70 (SAP_BASIS 770 — inline declarations, SWITCH, CONV,

string templates, @ host vars) **Development Account:** - `DEV-089` (Pass: `@Anhtuoi123`) — *Excel & Logic* - `DEV-061` (Pass: `@57Dt766`) — *SmartForms* - `DEV-242` (Pass: `12345678`) — *Email configuration*

Bước D1: Tạo Excel Template trên SMW0

Mục tiêu: Upload file Excel template chuẩn lên SAP server (SMW0).

Step 1: Tạo file Excel template trên máy local

Tạo file `ZTEMPLATE_PROJECT.xlsx` với các cột sau:

Column A	Column B	Column C	Column D	Column E	Column F	Column G
PROJECT_ID	**PROJECT_NAME**	**DESCRIPTION**	**START_DATE**	**END_DATE**	**PROJECT_MANAGER**	**NOTE**
(Char 20)	(Char 100)	(Char 255)	DD.MM.YYYY	DD.MM.YYYY	(Char 12)	(Char 255)
PRJ001	Example Proj	Description	01.04.2026	30.06.2026	DEV-089	

- **Row 1:** Headers (in đậm, có border)
- **Row 2:** Format hints / data types
- **Row 3:** Dữ liệu mẫu (highlight vàng)

Step 2: Upload lên SMW0

1. Vào T-code **SMW0**
2. Chọn **Binary data for WebRFC Applications**
3. Nhấn **Create** → nhập Object Name: `ZTEMPLATE_PROJECT`
4. Description: "Project Upload Template"
5. Chọn file `ZTEMPLATE_PROJECT.xlsx` từ local
6. **Save**

> Checkpoint: **SMW0** → `ZTEMPLATE_PROJECT` → **Download** → mở được file Excel chuẩn format.

Bước D2: Download Template Button

Mục tiêu: Nút trên GUI Status cho user download template.

Fcode: `DN_TMPL` trên `STATUS_0400` (Project List screen — initial screen trong flow mới).

> **Lưu ý flow mới:** Screen 0400 (Project List) là initial screen. Nút `DN_TMPL` và `UPLOAD` nằm trên `STATUS_0400`, chỉ visible cho Manager (xem `CODE_PBO.md` — non-Manager users exclude `UPLOAD` và `DN_TMPL`).

Trong `Z_BUG_WS_PAI`, handler đã có sẵn trong `user_command_0400`:

```
WHEN 'DN_TMPL'.
  PERFORM download_template.
```

FORM `download_template` (trong `Z_BUG_WS_F02`):

```
FORM download_template.
  DATA: lv_objid   TYPE wwwdatatab-objid VALUE 'ZTEMPLATE_PROJECT',
        lv_uaction TYPE i.

  " Chọn đường dẫn lưu file
  cl_gui_frontend_services=>file_save_dialog(
    EXPORTING
      default_extension = 'xlsx'
      default_file_name = 'ZTEMPLATE_PROJECT.xlsx'
      file_filter       = 'Excel Files (*.xlsx)|*.xlsx'
    CHANGING
      filename    = DATA(lv_filename)
      path        = DATA(lv_path)
      fullpath    = DATA(lv_fullpath)
      user_action = lv_uaction
    EXCEPTIONS OTHERS = 1 ).

  IF lv_uaction <> 0. " User cancelled
    MESSAGE s020(zbug_msg).
    RETURN.
  ENDIF.

  " Download từ SMW0 (primary method)
  DATA: lv_subrc TYPE sy-subrc.
  CALL FUNCTION 'DOWNLOAD_WEB_OBJECT'
    EXPORTING
      key         = VALUE wwwdatatab( relid = 'MI' objid = lv_objid )
      destination = lv_fullpath
    IMPORTING
      rc          = lv_subrc.

  IF lv_subrc = 0.
    MESSAGE s025(zbug_msg).  " 'Successfully saved'
  ELSE.
    " Fallback: SAP_OBJ_READ + GUI_DOWNLOAD (WWWDATA_IMPORT pattern)
    DATA: lt_mime TYPE TABLE OF w3mime,
          lv_size TYPE i.
    CALL FUNCTION 'SAP_OBJ_READ'
      EXPORTING
        p_objid   = lv_objid
        p_relid   = 'MI'
      TABLES
        p_content = lt_mime
```

```
      EXCEPTIONS
        not_found = 1
        OTHERS    = 2.

    IF sy-subrc = 0.
      CALL FUNCTION 'GUI_DOWNLOAD'
        EXPORTING
          filename = lv_fullpath
          filetype = 'BIN'
        TABLES
          data_tab = lt_mime
        EXCEPTIONS
          OTHERS   = 1.

      IF sy-subrc = 0.
        MESSAGE s025(zbug_msg).
      ELSE.
        MESSAGE s000(zbug_msg) WITH 'Failed to save' 'template file' DISPLAY LIKE
'E'.
      ENDIF.
    ELSE.
      MESSAGE s000(zbug_msg) WITH 'Template not found' 'on server (SMW0)' DISPLAY
LIKE 'E'.
    ENDIF.
  ENDIF.
ENDFORM.
```

Checkpoint: Bấm nút `DN_TMPL` trên Screen 0400 toolbar → chọn folder → file `.xlsx` lưu thành công.

Bước D3: Upload Excel Logic

Mục tiêu: Upload file Excel → validate → insert vào `ZBUG_PROJECT`.

Fcode: `UPLOAD` trên `STATUS_0400`.

Trong `Z_BUG_WS_PAI`, handler đã có sẵn trong `user_command_0400`:

```
WHEN 'UPLOAD'.
  PERFORM upload_project_excel.
```

FORM `upload_project_excel` (trong `Z_BUG_WS_F01`):

```
FORM upload_project_excel.
  DATA: lv_file     TYPE string,
        lt_raw      TYPE truxs_t_text_data,
        lt_projects TYPE TABLE OF zbug_project,
        ls_project  TYPE zbug_project,
        lv_errors   TYPE i,
```

```
      lv_success  TYPE i.

" 1. Chọn file
DATA: lt_file_table TYPE filetable, lv_rc TYPE i.
cl_gui_frontend_services=>file_open_dialog(
  EXPORTING file_filter = 'Excel Files (*.xlsx)|*.xlsx'
  CHANGING  file_table  = lt_file_table
            rc          = lv_rc
  EXCEPTIONS OTHERS = 1 ).

IF lv_rc <= 0. RETURN. ENDIF.
READ TABLE lt_file_table INTO DATA(ls_file) INDEX 1.
lv_file = ls_file-filename.

" 2. Đọc Excel vào internal table
TYPES: BEGIN OF ty_upload,
         project_id      TYPE char20,
         project_name    TYPE char100,
         description     TYPE char255,
         start_date      TYPE char10,
         end_date        TYPE char10,
         project_manager TYPE char12,
         note            TYPE char255,
       END OF ty_upload.
DATA: lt_upload TYPE TABLE OF ty_upload.

CALL FUNCTION 'TEXT_CONVERT_XLS_TO_SAP'
  EXPORTING
    i_field_seperator = 'X'
    i_line_header     = 'X'    " Bỏ qua header row
    i_tab_raw_data    = lt_raw
    i_filename        = lv_file
  TABLES
    i_tab_converted_data = lt_upload
  EXCEPTIONS
    conversion_failed   = 1
    OTHERS              = 2.

IF sy-subrc <> 0.
  MESSAGE s000(zbug_msg) WITH 'Failed to read' 'Excel file' DISPLAY LIKE 'E'.
  RETURN.
ENDIF.

" 3. Validate + Insert
LOOP AT lt_upload ASSIGNING FIELD-SYMBOL(<fs>).
  CLEAR ls_project.

  " Bỏ qua Header 2 dòng đầù (nêú i_line_header chưa clear hêt)
  IF <fs>-project_id CS 'PROJECT_ID' OR <fs>-project_id CS '(Char'.
```

```
  CONTINUE.
ENDIF.

" Validate PROJECT_ID không trống
IF <fs>-project_id IS INITIAL.
  ADD 1 TO lv_errors.
  CONTINUE.
ENDIF.

" Validate PROJECT_ID chưa tồn tại
SELECT COUNT(*) FROM zbug_project
  WHERE project_id = @<fs>-project_id.
IF sy-dbcnt > 0.
  ADD 1 TO lv_errors.
  CONTINUE.
ENDIF.

" Validate Project Manager exists + is Manager role + active
SELECT COUNT(*) FROM zbug_users
  WHERE user_id = @<fs>-project_manager AND role = 'M' AND is_del <> 'X'.
IF sy-dbcnt = 0.
  ADD 1 TO lv_errors.
  CONTINUE.
ENDIF.

" Map data to structure
ls_project-project_id      = <fs>-project_id.
ls_project-project_name    = <fs>-project_name.
ls_project-description     = <fs>-description.
ls_project-project_manager = <fs>-project_manager.
ls_project-note            = <fs>-note.

" Parse dates (DD.MM.YYYY → YYYYMMDD)
IF <fs>-start_date IS NOT INITIAL AND strlen( <fs>-start_date ) = 10.
  CONCATENATE <fs>-start_date+6(4) <fs>-start_date+3(2) <fs>-start_date(2)
    INTO ls_project-start_date.
ENDIF.
IF <fs>-end_date IS NOT INITIAL AND strlen( <fs>-end_date ) = 10.
  CONCATENATE <fs>-end_date+6(4) <fs>-end_date+3(2) <fs>-end_date(2)
    INTO ls_project-end_date.
ENDIF.

" Default values
ls_project-project_status = '1'.  " Opening
ls_project-ernam          = sy-uname.
ls_project-erdat          = sy-datum.
ls_project-erzet          = sy-uzeit.

APPEND ls_project TO lt_projects.
```

```
    ADD 1 TO lv_success.
  ENDLOOP.

  " 4. Batch insert
  IF lt_projects IS NOT INITIAL.
    INSERT zbug_project FROM TABLE lt_projects.
    COMMIT WORK.
    MESSAGE s000(zbug_msg) WITH 'Uploaded' lv_success 'project(s).' lv_errors.
    " Refresh project ALV after upload
    PERFORM select_project_data.
    PERFORM display_project_alv.
  ELSE.
    MESSAGE s000(zbug_msg) WITH 'No valid data' 'to upload' DISPLAY LIKE 'E'.
  ENDIF.
ENDFORM.
```

Checkpoint: Upload file Excel 3 rows:

- Row 1: Valid → Insert thành công
- Row 2: Duplicate `PROJECT_ID` → Skip
- Row 3: PM not Manager → Skip
- Message: "Uploaded 1 project(s). 2"
- Project ALV refreshes automatically

Bước D4: Message Class Migration

Mục tiêu: Thay tất cả hardcoded strings → `MESSAGE` class `ZBUG_MSG`.

TRƯỚC → SAU:

```
" TRƯỚC:
MESSAGE 'Chọn 1 dòng để cập nhật!' TYPE 'I'.
" SAU:
MESSAGE s009(zbug_msg).

" TRƯỚC:
MESSAGE 'Cập nhật thành công!' TYPE 'S'.
" SAU:
MESSAGE s025(zbug_msg).

" TRƯỚC (trong D2/D3):
MESSAGE 'Failed to read Excel file' TYPE 'S' DISPLAY LIKE 'E'.
" SAU:
MESSAGE s000(zbug_msg) WITH 'Failed to read' 'Excel file' DISPLAY LIKE 'E'.

" TRƯỚC:
MESSAGE 'Template downloaded successfully' TYPE 'S'.
```

```
" SAU:
MESSAGE s025(zbug_msg).
```

Quy tắc

1. Dùng `MESSAGE sXXX(zbug_msg)` cho popup messages (`S` = Success, `E` = Error)
2. Dùng `MESSAGE sXXX(zbug_msg) INTO lv_msg` cho internal messages
3. Dùng `DISPLAY LIKE 'E'` để hiện error icon dù type `S` (tránh dump màn hình)
4. Dùng `MESSAGE s000(zbug_msg) WITH '...' '...'` cho dynamic/fallback messages khi chưa có message number riêng (s000 = `&1 &2 &3 &4` generic placeholder)

Checkpoint: Chạy global search (Ctrl+F in SE80) `MESSAGE '` → không còn hardcoded message strings trong ZBUG_WORKSPACE_MP.

Bước D5: Orphan Bug Cleanup Script

Mục tiêu: Fix bugs cũ (tạo trước Phase A) không có `PROJECT_ID` — gán chúng vào 1 project hoặc đánh dấu.

Background: Trong flow mới, bug bắt buộc thuộc 1 project (`PROJECT_ID` enforce khi create). Nhưng bugs tạo từ hệ thống cũ có thể có `PROJECT_ID = ''`. Cần cleanup để tránh dữ liệu orphan.

Option A: Gán vào Default Project (Recommended)

Tạo report `Z_BUG_CLEANUP_ORPHAN` (SE38, Type 1 = Executable):

```
*&---------------------------------------------------------------------*
*& Report Z_BUG_CLEANUP_ORPHAN
*&---------------------------------------------------------------------*
*& Cleanup orphan bugs (no PROJECT_ID) — run ONCE after Phase A migration
*& Creates a default project "LEGACY" and assigns all orphan bugs to it
*&---------------------------------------------------------------------*
REPORT z_bug_cleanup_orphan.

CONSTANTS: lc_default_prj TYPE zde_project_id VALUE 'LEGACY'.

" ── Step 1: Check if default project exists, create if not ──
SELECT COUNT(*) FROM zbug_project
  WHERE project_id = @lc_default_prj.

IF sy-dbcnt = 0.
  DATA(ls_project) = VALUE zbug_project(
    project_id      = lc_default_prj
    project_name    = 'Legacy Bugs (Pre-Project)'
    description     = 'Auto-created to hold bugs from before Project entity was
introduced'
```

```
    project_status  = '1'   " Opening
    project_manager = sy-uname
    start_date      = sy-datum
    ernam           = sy-uname
    erdat           = sy-datum
    erzet           = sy-uzeit ).

  INSERT zbug_project FROM ls_project.
  IF sy-subrc = 0.
    WRITE: / 'Created default project:', lc_default_prj.
  ELSE.
    WRITE: / 'ERROR: Failed to create default project'.
    RETURN.
  ENDIF.
ELSE.
  WRITE: / 'Default project already exists:', lc_default_prj.
ENDIF.

" ── Step 2: Count orphan bugs ──
SELECT COUNT(*) FROM zbug_tracker
  WHERE project_id = @( CONV zde_project_id( '' ) )
    AND is_del <> 'X'.
DATA(lv_orphan_count) = sy-dbcnt.

WRITE: / 'Orphan bugs found:', lv_orphan_count.

IF lv_orphan_count = 0.
  WRITE: / 'Nothing to do — all bugs have a project.'.
  RETURN.
ENDIF.

" ── Step 3: Update orphan bugs ──
UPDATE zbug_tracker
  SET project_id = @lc_default_prj
      aenam      = @sy-uname
      aedat      = @sy-datum
      aezet      = @sy-uzeit
  WHERE project_id = @( CONV zde_project_id( '' ) )
    AND is_del <> 'X'.

IF sy-subrc = 0.
  COMMIT WORK.
  WRITE: / 'Successfully assigned', lv_orphan_count, 'bug(s) to project',
lc_default_prj.
ELSE.
  ROLLBACK WORK.
  WRITE: / 'ERROR: Update failed. Rolled back.'.
ENDIF.
```

```
" ── Step 4: Verify ──
SELECT COUNT(*) FROM zbug_tracker
  WHERE project_id = @( CONV zde_project_id( '' ) )
    AND is_del <> 'X'.
WRITE: / 'Remaining orphan bugs:', sy-dbcnt.
```

Option B: List-only (Dry Run)

Nếu chưa muốn gán ngay, chạy report này trước để xem danh sách orphan bugs:

```
REPORT z_bug_list_orphan.

SELECT bug_id, title, status, tester_id, dev_id, erdat
  FROM zbug_tracker
  WHERE project_id = @( CONV zde_project_id( '' ) )
    AND is_del <> 'X'
  INTO TABLE @DATA(lt_orphans).

IF lt_orphans IS INITIAL.
  WRITE: / 'No orphan bugs found.'.
ELSE.
  WRITE: / 'Orphan bugs (no PROJECT_ID):', lines( lt_orphans ).
  SKIP.
  LOOP AT lt_orphans ASSIGNING FIELD-SYMBOL(<o>).
    WRITE: / <o>-bug_id, <o>-title, <o>-status, <o>-tester_id, <o>-dev_id, <o>-
erdat.
  ENDLOOP.
ENDIF.
```

Khi nào chạy?

- Chạy **1 lần duy nhất** sau khi Phase A hoàn tất (tables đã có `PROJECT_ID` field)
- Chạy **trước** Phase C go-live (trước khi users bắt đầu dùng Project-first flow)
- Sau khi chạy, verify: `SELECT COUNT(*) FROM zbug_tracker WHERE project_id = '' AND is_del <> 'X'` → phải = 0

Checkpoint: Chạy `Z_BUG_CLEANUP_ORPHAN` (hoặc list trước rồi cleanup) → tất cả bugs đều có `PROJECT_ID` → Project ALV → click LEGACY → thấy tất cả bugs cũ.

TỔNG KẾT PHASE D

Sau khi hoàn thành Phase D, bạn phải có:

- ☐ Excel template `ZTEMPLATE_PROJECT` trên SMW0
- ☐ Download Template button (`DN_TMPL`) hoạt động trên Screen 0400 (với fallback SAP_OBJ_READ)
- ☐ Upload Excel (`UPLOAD`) → validate PM role + duplicate check + date parsing → insert `ZBUG_PROJECT` + auto-refresh ALV

- ⍰ Message Class `ZBUG_MSG` dùng xuyên suốt (không còn hardcode)
- ⍰ Orphan bugs cleanup — tất cả bugs có `PROJECT_ID` (no orphans)

Lưu ý: Dashboard Statistics (D5 trong phiên bản cũ) đã **cancelled** — Screen 0100 (Homepage) deprecated trong flow mới. Nếu cần statistics sau này, có thể implement trực tiếp trên Screen 0400 toolbar hoặc popup report riêng.

Chuyển sang **Phase E: Testing & Go-Live** # HƯỚNG DẪN TRIỂN KHAI CHI TIẾT — PHASE E: TESTING & GO-LIVE

Dự án: SAP Bug Tracking Management System **Ngày:** 24/03/2026 | **Phiên bản:** 5.0 (Module Pool Integration) **Thời gian ước tính:** 2 ngày (02-03/04) **Yêu cầu:** Hoàn thành Phase A + B + C + D trước **Development Account:** - `DEV-118` (Pass: `Qwer123@`) — *Testing & Final Review*

Bước E1: Cập nhật T-Code ZBUG_WS + Deprecate T-Codes cũ

E1.1: Chuyển T-code ZBUG_WS sang Module Pool

Mục tiêu: Chuyển T-code `ZBUG_WS` từ `Z_BUG_WORKSPACE` (SE38) sang `Z_BUG_WORKSPACE_MP` (Module Pool).

Vào **SE93** → nhập `ZBUG_WS` → **Change** (hoặc Delete/Create mới).

1. Tạo `ZBUG_WS`
2. Chọn **Dialog transaction**
3. **Program:** `Z_BUG_WORKSPACE_MP`
4. **Screen number:** `0100` (main hub)
5. Lựa chọn **GUI support** (Windows/Web GUI)
6. **Save**

> ✅ **Checkpoint:** Gõ `/nZBUG_WS` → mở ngay cửa sổ Bug Tracking Workspace (Module Pool).

E1.2: Deprecate T-Codes cũ

Mục tiêu: Đánh dấu các T-codes cũ (SE38) là deprecated, chưa xóa — giữ làm fallback cho đến khi go-live ổn định.

Vào **SE93** → lần lượt mở từng T-code → **Change** → sửa **Description** thêm prefix `[DEPRECATED]:`

T-Code	Description mới	Program cũ
`ZBUG_CREATE`	`[DEPRECATED] Use ZBUG_WS - Create Bug`	`Z_BUG_CREATE_SCREEN`
`ZBUG_UPDATE`	`[DEPRECATED] Use ZBUG_WS - Update Bug`	`Z_BUG_UPDATE_SCREEN`
`ZBUG_REPORT`	`[DEPRECATED] Use ZBUG_WS - Bug Report`	`Z_BUG_REPORT_ALV`
`ZBUG_MANAGER`	`[DEPRECATED] Use ZBUG_WS - Manager Dashboard`	`Z_BUG_MANAGER_DASHBOARD`
`ZBUG_PRINT`	`[DEPRECATED] Use ZBUG_WS - Print Bug`	`Z_BUG_PRINT`
`ZBUG_USERS`	`[DEPRECATED] Use ZBUG_WS - User Management`	`Z_BUG_USER_MANAGEMENT`

⚠️ **Lưu ý:** KHÔNG xóa T-codes cũ. Chỉ xóa sau khi E6 testing confirm Module Pool hoạt động ổn định và go-live thành công.

✅ **Checkpoint:** Gõ `/nZBUG_CREATE` → vẫn mở được (fallback), nhưng description hiện `[DEPRECATED]`.

Bước E2: Unit Test — Function Modules

Mục tiêu: Test từng FM bằng SE37, đảm bảo data manipulation chuẩn xác.

Test Case 01: `Z_BUG_CHECK_PERMISSION`

- **TC-01a:** `IV_USER` = 'DEV-Manager', `IV_ACTION` = `CREATE_PROJECT` 👉 **Expected:** `EV_ALLOWED` = `Y`
- **TC-01b:** `IV_USER` = 'DEV-Tester', `IV_ACTION` = `CREATE_PROJECT` 👉 **Expected:** `EV_ALLOWED` = `N`, `EV_MESSAGE` = "Only Manager…"

Test Case 02: `Z_BUG_CREATE`

- **TC-02a:** Basic creation. 👉 **Expected:** `EV_SUCCESS` = `Y`, Return Bug ID hợp lệ.
- **TC-02b:** Severity=1 (Dump), Bug Type=C. 👉 **Expected:** Priority force thành `H` (High).

Test Case 03: `Z_BUG_UPDATE_STATUS`

- **TC-03a:** Valid transition: 1 → 2. 👉 **Expected:** Success.
- **TC-03b:** Invalid transition: 1 → 5. 👉 **Expected:** `EV_SUCCESS` = `N`.

Test Case 04: `Z_BUG_AUTO_ASSIGN`

- **TC-04a:** Create Code Bug trong project có 1 active dev (role='D', is_del<>'X', thuộc project). 👉 **Expected:** Bug tự động STATUS=2, DEV_ID = tên dev đó.
- **TC-04b:** Create Code Bug trong project có 0 active dev. 👉 **Expected:** STATUS = 'W' (Waiting), DEV_ID = blank.

Test Case 05: `Z_BUG_REASSIGN`

- **TC-05a:** Manager reassign bug từ Dev1 sang Dev2 (cùng project). 👉 **Expected:** `EV_SUCCESS` = `Y`, DEV_ID = Dev2, STATUS = '2'.
- **TC-05b:** Manager reassign sang dev KHÔNG thuộc project. 👉 **Expected:** `EV_SUCCESS` = `N`, "User is not a member of this project".

> ✅ **Checkpoint:** Tất cả test cases trên SE37 phải PASS (EV_SUCCESS = 'Y' cho valid cases).

Bước E3: Integration Test — Full Workflow

Mục tiêu: Test toàn diện end-to-end user journey qua UI Module Pool.

Workflow 01: Code Bug (Happy Path)

1. Logon Tester → **/nZBUG_WS** → Create Code Bug (Sev=3, Prio=H).
2. STATUS=1 (New), DEV_ID = *Blank*.
3. Auto-Assign hoặc Manager manually assign → STATUS=2 (Assigned to Dev).
4. Logon Dev → Bug Detail → Update Status: 2 → 3 (In Progress).
5. Dev upload *TESTCASE* (GOS).
6. Dev update Status: 3 → 5 (Fixed).
7. Logon Tester → Check *TESTCASE*, upload *CONFIRM* (GOS), 5 → 6 (Resolved).
8. Logon Manager → Close Bug (6 → 7).
9. Mở **History Tab**: Phải có log CR, AS, IP, FX, RS, CL.
10. Check **SOST**: 3~4 emails đã gửi trong quá trình đổi trạng thái.

Workflow 02: Config Bug (Tester Self-Fix)

1. Tester Create Bug: Type=F (Config).
2. Bug tự động chuyển STATUS=2, DEV_ID = Tên Tester.
3. Tester tự sửa → Đẩy trạng thái 2 → 3 → 5 → 6.
4. Manager close it.

Workflow 03: Reject + Reassign Path

1. Tester tạo Code Bug → STATUS=1 (New).
2. Auto-assign → STATUS=2, DEV_ID = Dev1.
3. Dev1 reject → STATUS=R (Rejected).

4. Manager reassign sang Dev2 → STATUS=2, DEV_ID = Dev2.
5. Dev2 bắt đầu → STATUS=3 (InProgress).
6. Dev2 fix xong → STATUS=5 (Fixed), upload TESTCASE.
7. Tester verify pass → STATUS=6 (Resolved), upload CONFIRM.
8. Manager close → STATUS=7 (Closed).
9. Mở **History Tab**: Phải có log CR, AS, RJ, RS (Reassign), IP, FX, RS (Resolved), CL.

Workflow 04: Pending Path

1. Bug đang ở STATUS=3 (InProgress).
2. Dev set Pending (chờ thông tin thêm) → STATUS=4 (Pending).
3. Dev resume khi có thông tin → STATUS=3 (InProgress).
4. Dev fix → STATUS=5 → Tester verify → STATUS=6 → Manager close → STATUS=7.
5. Mở **History Tab**: Phải có log chuyển 3→4, 4→3.

✅ **Checkpoint:** Cả 4 luồng đều hoàn tất End-to-end không phát sinh ABAP Dump.

Bước E4: Permission Matrix Test

Mục tiêu: Áp đặt đúng quyền cho từng Role.

Action	Tester (T)	Developer (D)	Manager (M)
CREATE Bug	✅	❌	✅
UPDATE_STATUS	✅ (*1)	✅ (*2)	✅
DELETE Bug	❌	❌	✅
UPLOAD_REPORT	✅ (*3)	❌	✅
UPLOAD_FIX	✅ (*4)	✅ (*2)	✅
CREATE/CHANGE PRJ	❌	❌	✅

Chú thích:

- (*1): Mới tạo Status=1 hoặc Config bug do mình giữ.
- (*2): Chỉ bug đang được giao (DEV_ID).
- (*3): Chỉ bug do chính Tester tạo ra.
- (*4): Config bug do chính Tester giữ.

✅ **Checkpoint:** Thử nghiệm sai Role → Lỗi “Not authorized to update this bug” (s006).

Bước E5: Feature-Specific Tests

E5.1: GOS File Upload

Upload 1 excel + 1 pdf. Chuyển sang Evidence Tab (0350) hoặc GOS Component phải liệt kê đủ 2 file.

E5.2: SmartForm Email

Nội dung mail phải ở dạng HTML (không phải Plain Text thô bạo), đầy đủ field. Check **SOST** để xác nhận.

E5.3: Đa Ngôn Ngữ

Logout SAP, login lại Language = *VI*. Click các component, xem message pop-up phải ra Tiếng Việt.

E5.4: Long Text (Dev Note / Tester Note / Root Cause)

1. Mở Bug Detail → Tab **Developer Note** (0320).
2. Nhập text dài (nhiều dòng, có ký tự đặc biệt).
3. **Save** → Đóng Bug Detail → Mở lại.
4. Text phải persist đúng nguyên vẹn (qua `READ_TEXT`).
5. Check **SE75** → Text Object `ZBUG` → Text ID `Z001` → nhập Bug ID → xem text đã lưu.
6. Lặp lại cho Tab **Tester Note** (0330, Text ID `Z002`) và **Root Cause** (0340, Text ID `Z003`).

✅ **Checkpoint:** Long Text lưu và đọc lại chính xác, không mất dữ liệu.

E5.5: Status Migration Verification

Mục tiêu: Xác nhận report `Z_BUG_MIGRATE_STATUS` (Phase A, Bước A8) đã chạy đúng.

1. **SE16** → `ZBUG_TRACKER`:
   - Không còn record nào có `STATUS = '4'` mà KHÔNG phải Pending (kiểm tra các bug cũ đã migrate từ Fixed→5).
   - Không còn record nào có `STATUS = '6'` theo nghĩa cũ (Deleted) — tất cả đã có `IS_DEL = 'X'`.
   - Tất cả bug cũ có `STATUS = '5'` (old Closed) giờ hiện `STATUS = '7'`.
2. **SE16** → `ZBUG_HISTORY`:
   - Kiểm tra old_value/new_value đã được remap tương ứng (4→5, 5→7).

✅ **Checkpoint:** Không còn status code cũ nào conflict với 9-state model mới.

Bước E6: Clean Test Data + Demo Day Rehearsal

Step 1: Clean test data (`SE38`)

```
DELETE FROM zbug_tracker WHERE bug_id LIKE 'BUG9%'.
DELETE FROM zbug_project WHERE project_id LIKE 'TEST%'.
DELETE FROM zbug_history WHERE bug_id LIKE 'BUG9%'.
DELETE FROM zbug_user_projec WHERE project_id LIKE 'TEST%'.
COMMIT WORK.
```

Step 2: Demo Day Script (03/04/2026)

- (2 phút) Giới thiệu hệ thống, tổng quan kiến trúc Module Pool.
- (3 phút) Login Manager → Tạo project, Import Users.
- (3 phút) Login Tester → Tạo Code Bug → Mở form validation.
- (2 phút) Auto-assign chạy → Check mail SOST cho Demo.
- (3 phút) Developer fix bug, upload file đính kèm BDS (Evidence Tab).
- (1 phút) Show Reject + Reassign flow (Dev reject → Manager reassign).
- (1 phút) Show Long Text notes (Dev Note, Tester Note tabs).
- (1 phút) Show UI Tab Strip 6 tabs, History logging tự động.
- (1 phút) Show Excel Upload & Validation (Project Upload).
- (1 phút) Show Pending flow (3→4→3).

Step 3: Transport Request

1. Gộp toàn bộ object vào 1 Transport Request (TR).
2. Object list: Tables, Domains, DEs, Programs, Includes, FMs, SmartForms, Texts…
3. Release TR (SE10).

🎯 **ĐÍCH ĐẾN CUỐI CÙNG: 03/04/2026 GO-LIVE THÀNH CÔNG.**